

BASF

We create chemistry

NeoHelix® Regenerate

The smart peptide for next-level skin

Global Except US

TABLE OF CONTENTS

Summary file 3

From structure to signaling: collagen's essential functions in skin 4

Collagen degradation: from healthy architecture to structural dysfunction 5

Collagen hybridizing peptides (CHPs): structure based targeting of damaged collagen ... 7

NeoHelix Regenerate: designed for targeted collagen repair and skin regeneration 8

Demonstrated efficacy 14

Epidermal regeneration booster (*ex vivo*) 15

Decreased release of pro-inflammatory cytokines by keratinocytes (*in vitro*) 18

Boosted collagen production by fibroblasts (*in vitro*) 19

Matrix protection from glycation (*ex vivo*) 21

Boosted leptin in adipocytes (*ex vivo*) 22

Acting on dermal renewal: repairing collagen, boosting hyaluronic acid (*in vivo*) 23

Improvement of skin aging: reducing wrinkles increasing skin tonicity and thickness (*in vivo*) 27

General conclusion 31

Annexes 32

Bibliography 33

The trademarks symbolized with a ® or ™ are either property of or licensed to BASF group and registered and/or applied for registration in relevant countries. Other product names and trademarks mentioned may belong to third parties.

SUMMARY FILE

NeoHelix® Regenerate	BC10226
Origin - Description	Synthetic biomimetic Collagen Hybridizing Peptide designed to bind damaged collagen and promote skin repair
Regulatory data	
INCI	Mannitol (and) Acetyl Tetra (Tripeptide-29) Dipeptide-73 Hydroxyproline Tripeptide-122 Tetra (Tripeptide-29)
China	Acetyl Tetra (Tripeptide-29) Dipeptide-73 Hydroxyproline Tripeptide-122 Tetra (Tripeptide-29) is NOT listed in Inventory of Existing Cosmetic Ingredient in China (IECIC)
CAS#	69-65-8; 3099375-39-7
Appearance	White to off white powder with characteristic odor
Natural labels	None
Naturalness content (ISO 16128)	99.9% from natural origin
Cosmetic use	
Properties	• Specifically binds to damaged collagen • Supports epidermal (*ex vivo*) and dermal (*in vitro*) regeneration • Helps protect skin proteins from glycation (*ex vivo*) • Helps reduce damaged collagen in the skin • Helps reduce the appearance of wrinkles • Helps improve skin tonicity and thickness
Applications	• Anti-wrinkle / Anti-aging facial care • Pre- and post-aesthetic procedure facial care
Formulation data	
Concentration of use	1%
Solubility	Soluble in water, insoluble in oils and fats
Incorporation method	Incorporate into the cosmetic product up to 75°C or at room temperature for cold processing
Optimal pH	4-8



From structure to signaling: collagen's essential functions in skin

Collagen is the most abundant protein in the skin, representing about 75% of its dry weight [Salamito *et al*, 2025; Jadach *et al*, 2024]. It is primarily located in the dermis, where it forms a dense fibrous network that provides mechanical strength and elasticity.

Collagen architecture is defined by a triple-helix configuration, formed by three polypeptide chains tightly wound together. This unique structure provides exceptional tensile strength and elasticity, essential for maintaining skin firmness and resilience.

Each collagen chain consists mainly of repeating (Gly–X–Y) triplets, where glycine (Gly) occupies every third position to allow tight packing of the helix, while X and Y are typically proline and hydroxyproline, which stabilize the structure through hydrogen bonding [Kanniyappan *et al*, 2025; Emam, 2025].

Beyond the molecular level, collagen triple helices self-assemble into microfibrils, which further aggregate to form fibrils, the fundamental structural units of the dermal extracellular matrix (ECM). These fibrils then bundle together into collagen fibers, creating a highly organized and resilient scaffold that supports skin architecture (Fig. 1). The hierarchical organization, from triple helix to microfibril, fibril, and finally fiber plays a crucial role in determining the skin's mechanical properties, including tensile strength, elasticity, and resistance to deformation.

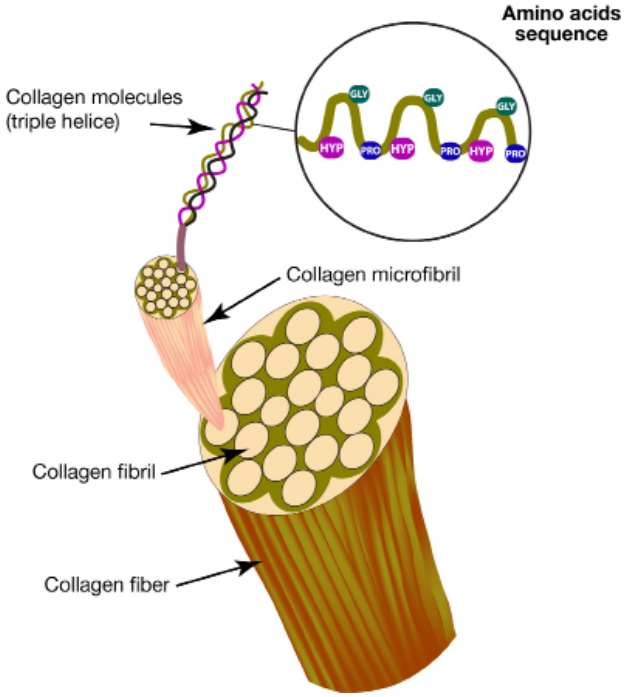


Figure 1 - Collagen structure.

In the dermis, various collagen types contribute to tissue **architecture**, notably Type I and Type III, the most abundant fibrillar collagens [Zhou *et al*, 2024], while additional minor types, such as collagen IV, V and VII, ensure structural cohesion and proper dermo-epidermal adhesion [Tohgasaki *et al*, 2022; Queiroz *et al*, 2025].

Beyond its mechanical contribution to dermal architecture, collagen acts as an essential regulator of **cell–matrix communication**. Collagen binding integrins expressed on the surface of fibroblasts, including integrins $\alpha1\beta1$, $\alpha2\beta1$, $\alpha10\beta1$ and $\alpha11\beta1$, recognize specific triple-helical motifs such as GFOGER (Gly Phe Hyp Gly Glu Arg) [Emsley *et al*, 2000; Malcor *et al*, 2022]. This integrin mediated recognition enables fibroblasts to adhere to collagenous matrices and to sense the biochemical and structural properties of the extracellular matrix [Khew *et al*, 2008].

Notably, the GFOGER motif appears only once within the collagen molecule and is found exclusively in collagen types I and IV [Knight *et al*, 2000]. More specifically, in Type I collagen, GFOGER peptide sequence corresponds to residues 502–507 of the $\alpha1$(I) chain [Liu *et al*, 2010].

The functional recognition of GFOGER requires its correct three-dimensional presentation: several studies have shown that this peptide sequence must be presented in a structured conformation, such as within a stabilized triple helix, to ensure effective binding to integrins [Knight *et al*, 2000; Emsley *et al*, 2004; Jokinen *et al*, 2010]. This high-affinity interaction orchestrates key fibroblast-driven processes, including adhesion, migration, cytoskeletal organization, and signaling events, thereby contributing to tissue remodeling and repair, where an increase in extracellular matrix synthesis is expected to be reactivated [Pang *et al*, 2023; Hynes, 2002].

Collagen exhibits a dual structural and bioactive nature maintaining dermal homeostasis: its fibrillar framework delivers essential mechanical stability to the dermal matrix, while its biochemical signaling capabilities regulate fibroblast function and orchestrate extracellular matrix remodeling.

Collagen degradation: from healthy architecture to structural dysfunction

A healthy collagen network not only maintains the structural architecture of the skin, but also acts as a silent messenger, signaling cells to produce the components required to preserve a youthful appearance. When this support system deteriorates, the skin loses firmness, and early wrinkles start to form.

Unfortunately, our collagen is subjected every day to micro aggressions that gradually impair its structure. Key contributors include:

- **UV radiation**

UV rays, particularly UVA, penetrate the dermis and generate free radicals. These reactive species activate several matrix metalloproteinases (MMPs), notably MMP-1, MMP-3 and MMP-9, which act like molecular scissors, cleaving the structural collagen network [Feng *et al*, 2024]. As these enzymes break down collagen I and III, the extracellular matrix becomes progressively fragmented and disorganized, weakening the dermal scaffold that supports fibroblasts. In this altered environment, fibroblasts lose their optimal mechanical support, adopt a dysfunctional phenotype, and produce less collagen. The combined effect of collagen degradation, matrix fragmentation, and fibroblast impairment accelerates dermal thinning and leads to the formation of premature wrinkles [Quan *et al*, 2013].

- **Atmospheric pollution**

Urban pollutants and fine particles induce oxidative stress within skin cells. Exposed cells upregulate MMP-1 and MMP-3, accelerating the breakdown of collagen [Martic *et al*, 2022]. Living in polluted urban environments is therefore associated with dullness, loss of firmness, and more pronounced wrinkles [Schikowski *et al*, 2020].

- **Oxidative stress**

Chronic oxidative stress does not act in isolation, it triggers a persistent low grade inflammatory state, which further amplifies the skin aging cascade. Chronic inflammation is now recognized as a major “hallmark” of skin aging, directly contributing to the upregulation of MMP-1, one of the key collagen degrading enzymes responsible for dermal structural decline [Lai *et al*, 2026].

In addition, in photodamaged or ROS (reactive oxygen species)–rich environments, inflammatory mediators activate signaling pathways such as MAPK/ERK, that stimulate AP-1, a transcription factor that increases MMPs expression and accelerates collagen breakdown [Feng *et al*, 2024]. Over time, uncontrolled oxidative stress speeds up collagen degradation and reduces new collagen synthesis, creating a self perpetuating aging cycle [Tu *et al*, 2016].

- **Glycation**

Glycation is a slow chemical reaction in which circulating sugars bind to collagen fibers, forming advanced glycation end products (AGEs). These irreversible crosslinks stiffen collagen, compromising its function. Even worse, glycated collagen becomes resistant to enzymatic repair, causing damaged fibers to accumulate because the body can no longer break them down effectively. The accumulation rate of glycated collagen is estimated to increase by about 3.7% per year, although lifestyle, UV exposure and diet can accelerate this percentage [Danby, 2010]. As collagen becomes increasingly crosslinked, these rigidified fibers interact less efficiently with surrounding cells, altering mechano-signaling and disrupting dermal homeostasis [Kasper *et al*, 2001]. In addition, AGEs are not merely passive structural defects: they actively enhance the expression and activity of several MMPs, including MMP-1, MMP-2, and MMP-9 [Mahmoud *et al*, 2021; Han *et al*, 2016] and trigger an increase in intracellular ROS [Zheng *et al*, 2022]. This oxidative stress further amplifies MMPs production and accelerates collagen degradation, compounding the overall matrix damaging effects.

Together, these stresses (UV, pollution, oxidative stress and glycation) amplify extracellular matrix breakdown and accelerate visible and functional skin aging.

Damaged collagen not only disrupts skin structure but also collapses the fibroblast communication system: fibroblasts receive weakened signals, reducing their ability to regenerate the extracellular matrix. They shift toward a dysfunctional state marked by decreased collagen synthesis and increased MMPs activity, which accelerates matrix breakdown. **This creates a vicious cycle** in which collagen damage further **reduces fibroblast stimulation, amplifies enzymatic degradation**, and progressively **undermines the dermis’s regenerative capacity and structural integrity** [Vanari *et al*, 2006; Tu *et al*, 2016; Fisher *et al*, 2023].

In addition to these impacts on the dermis, collagen damage exerts **far-reaching effects on all levels of skin architecture**:

- A compromised extracellular matrix indirectly slows epidermal renewal, as keratinocyte proliferation and differentiation depend on signals and structural support provided by a stable, functional dermal matrix. An organized collagen network is essential to maintain epidermal homeostasis: when the extracellular matrix remains intact, it supports efficient keratinocyte growth, stratification, and barrier formation, whereas matrix disorganization compromises overall epidermal quality [Huang *et al*, 2022; Potekaev *et al*, 2021; Watt *et al*, 2011]. **As a result, the epidermis becomes thinner, less resilient, and less capable of effective repair**.
- It also has an impact on the hypodermis as fibroblasts and adipocytes communicate **through reciprocal signaling** that coordinates skin structure and function across dermal and hypodermal layers. Dermal fibroblasts regulate adipocyte differentiation by secreting factors such as TGFβ, thereby influencing tissue remodeling [Sander *et al*, 2024]. Conversely, adipocytes release adipokines, including leptin and adiponectin, which modulate fibroblast activation and extracellular matrix production [Li *et al*, 2005; Ezure *et al*, 2007]. Alterations in this dialogue, particularly those arising from collagen damage and fibroblast dysfunction, lead to **impaired dermo-hypodermal communication and age-related skin decline**.

Collagen degradation initiates a cascade of structural and signaling dysfunctions that extend across all skin layers. By disrupting extracellular matrix integrity, damaged collagen impairs fibroblast function, alters epidermal renewal, and weakens dermo-hypodermal communication. This progressive loss of intercellular coordination ultimately compromises global skin homeostasis and accelerates visible and functional skin aging.

Collagen Hybridizing Peptides (CHPs): structure-based targeting of damaged collagen

All mammalian collagens share a highly conserved **Gly–X–Y repeating sequence**, which governs the assembly of three collagen α-chains into a tightly packed triple-helical structure. Structural damages caused by thermal stress, enzymatic degradation (notably by matrix metalloproteinases induced by UV, oxidative stress, pollution and glycation), or mechanical injury lead to partial or complete unwinding of this triple helix, exposing individual collagen α-chains that are normally inaccessible in healthy tissues [Shoulders *et al*, 2009; Leikina *et al*, 2002; Fields, 2013].

Derived from biomedical science, backed by over 650 scientific publications, the **Collagen Hybridizing Peptide** (CHP) technology is based on peptide sequences which specifically recognize and bind these damaged collagen chains through a **triple-helix forming mechanism** (Fig. 2) [Li *et al*, 2023].

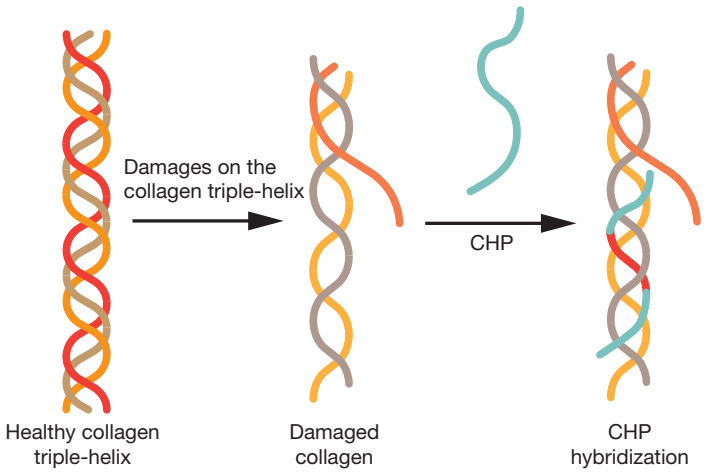


Figure 2 - Schematic illustration of a CHP binding to an unfolded collagen strand.

CHP binding relies on a **structure-guided hybridization mechanism** in which the peptide selectively recognizes exposed collagen chains. Upon interaction with unwound collagen, CHPs spontaneously assemble into collagen-CHP heterotrimeric triple helical complexes through heterogeneous triple-helix formation. This hybridization is stabilized by extensive backbone hydrogen bonding and closely mimics the molecular organization of native collagen triple-helices. The high specificity of this interaction is driven by the neutral, hydrophilic and proline-rich sequence composition of CHPs, which promotes triple-helix compatibility while minimizing non-specific interactions [Hwang *et al*, 2017; Bennink *et al*, 2018; Li *et al*, 2023]. **CHPs exhibit high affinity for denatured collagen by selectively hybridizing with exposed collagen α-chains in partially unfolded triple helices, while showing no detectable binding to intact, fully folded triple-helical collagen in healthy tissues or to non collagenous extracellular matrix proteins [Li *et al*, 2013].**

As all collagen subtypes contain triple-helical regions and because collagen structure is conserved across species, CHPs are not species-specific and can bind to **all 28 types of collagens** [Li *et al*, 2023].

CHP technology is widely used in the **biomedical field** for histopathology (detection of myocardial infarction, arthritis, nephritis and fibrosis) and for identifying mechanical collagen damage across connective tissues such as bone, tendon, ligament, heart valves and blood vessels [Krishnamoorthy *et al*, 2018; Converse *et al*, 2018, Hwang *et al*, 2017].

In the skin, CHP technology has successfully been used to detect collagen damage in **thermally injured human tissue** [Schroeder *et al*, 2020; Kwa *et al*, 2020] and during **wound healing**, where collagen remodeling is actively occurring [Hwang *et al*, 2017].

In vivo experiments have shown CHPs association with damaged collagen for up to 5 days, consistent with sustained detection following systemic administration [Linder *et al*, 2024].

CHP is a highly selective and structure-guided technology to target damaged collagen through direct recognition of denatured collagen chains.

NeoHelix Regenerate: designed for targeted collagen repair and skin regeneration

NeoHelix Regenerate merges CHP technology with an integrin-binding sequence to drive next-generation skin regeneration. Precisely engineered to target damaged collagen, it selectively anchors to sites where repair is required. Simultaneously, integrin engagement enhances fibroblast-matrix interactions, supporting advanced mechanisms of skin regeneration.

Characteristics of NeoHelix Regenerate

A new peptide class

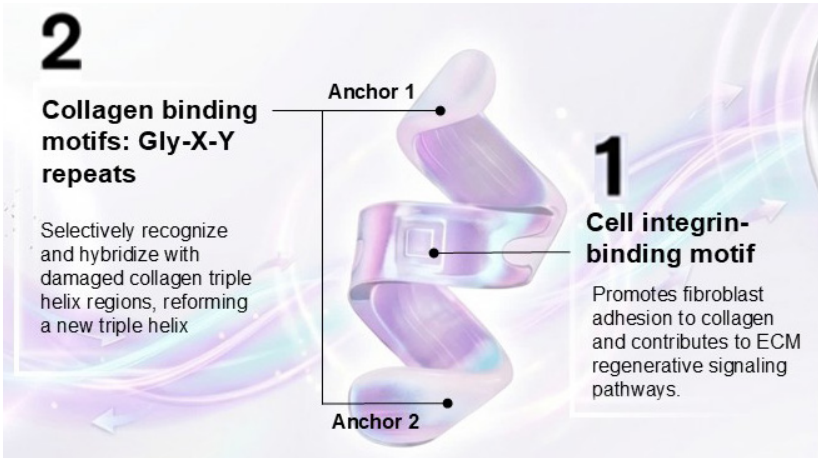


Figure 3 - Schematic illustration of the NeoHelix Regenerate peptide structure.

Its structure incorporates two collagen-binding motifs based on Gly-X-Y repeats, predominantly composed of glycine and proline, which mimic native collagen sequences. These collagen-mimetic domains selectively recognize and hybridize with damaged collagen. In addition, the peptide contains an integrin-binding motif (GFOGER) contributing to the activation of matrix-regenerative signaling pathways and reinforcing cell-matrix interactions (Fig. 3).

Figure 4 schematically illustrates the mechanism of action of NeoHelix Regenerate. After penetrating the skin and reaching the dermis, the NeoHelix Regenerate peptide selectively binds to damaged collagen. This selective interaction allows the peptide to anchor at sites of matrix damage. Once anchored, its integrin-binding motif promotes fibroblast adhesion to collagen and reactivates integrin-mediated signaling pathways. This leads to fibroblast reactivation, supporting extracellular matrix renewal and stimulating collagen synthesis, thereby contributing to skin regeneration.

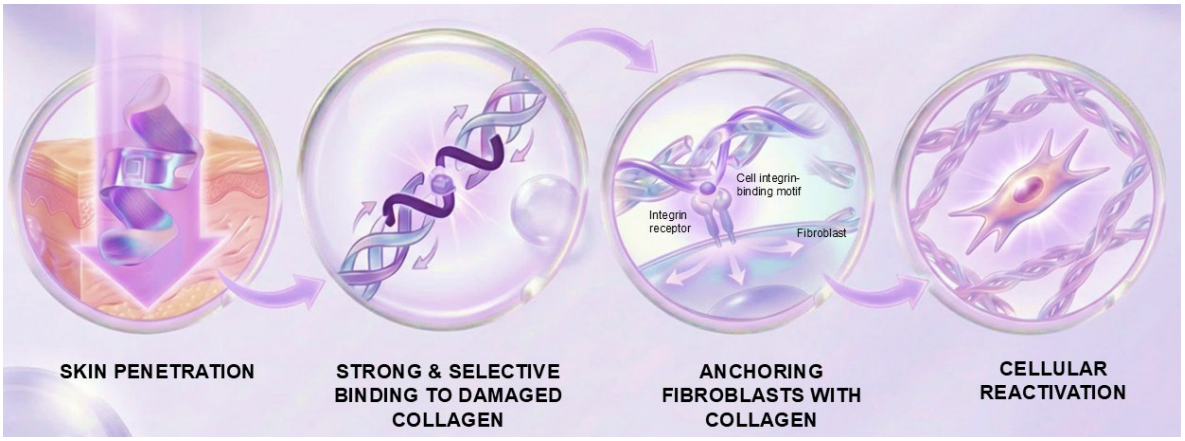


Figure 4 - Illustration of the mechanism of action of NeoHelix Regenerate.

Peptide INCI: Acetyl Tetra (Tripeptide-29) Dipeptide-73 Hydroxyproline Tripeptide-122 Tetra (Tripeptide-29).

Homology with collagen: NeoHelix Regenerate peptide sequence is extremely close to human collagen type I, as it contains Gly-X-Y repeats and the integrin-binding sequence Gly-Phe-Hyp-Gly-Glu-Arg (GFOGER), both 100% identical to specific regions of the human COL1A1 α–1 chain (sequence alignments done with Clustal Omega tool from the EMBL-EBI Job Dispatcher sequence analysis tools framework platform PMID: 38597606 [Madeira *et al*, 2024]). The domains Gly-X-Y correspond to positions 142-153 and 1177-1189 in the α-1 chain of human collagen type I (COL1A1), while the integrin-binding sequence aligns with positions 680-685 (proline was used instead of hydroxyproline in the alignment because hydroxyproline results from a post translational modification).

Number of amino acids: 30

Molecular weight: 2730 g/mol

Melting temperature (Tm): ~32°C , measured by circular dichroism.

T_m is defined as the temperature at which the peptides transition from a triple-helix to an unfolded structure [Ahmad *et al*, 2025]. The measured Tm of NeoHelix Regenerate lies well above physiological and cosmetic use temperatures, confirming the suitability of this peptide for cosmetic applications.

Solubility: Water soluble.

Heat stability: stable in water-based solution after 6 hours of heating at 75°C.

A smart precision binding peptide

Able to reach the dermis

A 24-hour *ex vivo* study performed on human skin demonstrates, by fluorescence microscopy, the penetration and localization of the fluorescein-tagged NeoHelix Regenerate peptide within the dermal matrix (Fig. 5). This result shows that the peptide can reach its biological target in the dermis. In addition, remanence studies performed on human skin biopsies demonstrated detectable NeoHelix Regenerate binding for up to three days (data not shown), indicating that the peptide may persist long enough within the skin to ensure sustained interaction with damaged collagen and regenerative activity.

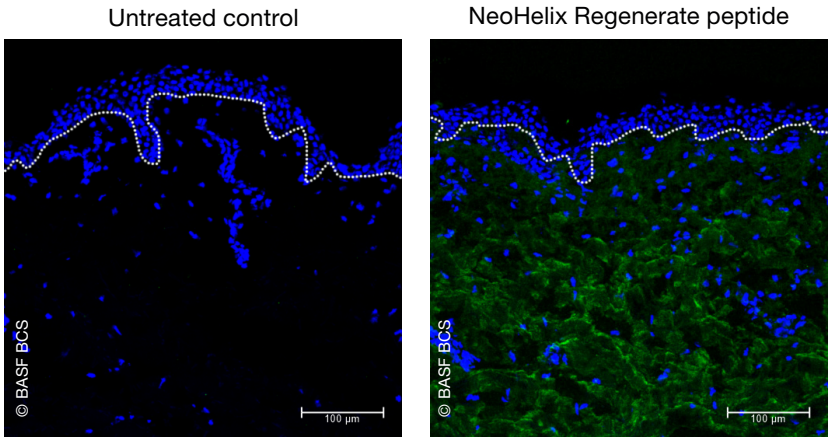


Figure 5 - Representative images illustrating skin penetration of NeoHelix Regenerate peptide 24h post topical treatments (in green fluorescein-tagged NeoHelix Regenerate peptide; in blue: DAPI-stained nuclei; the dermo-epidermal junction is represented by a dotted line).

MATERIALS&METHODS

Study number
EMB230801-2024-089-HI

Tested products
Fluorescein-tagged NeoHelix Regenerate peptide (MOL_0001130_001) was tested diluted in PBS (Phosphate-Buffered Saline) at a concentration of 150 µM and stored at room temperature.

Treatment
20 µL of Fluorescein-tagged NeoHelix Regenerate peptide solution at 150 µM was topically applied to a human skin explant (Genoskin; female donor, 37 years old). The solution was uniformly distributed over the explant surface using a glass rod for 30 seconds. A second topical application was performed 6 hours after the initial treatment. In parallel, an untreated skin explant was included. The experiment stopped after 24 hours.

Penetration visualization
The skin explant was embedded in optimal cutting temperature (OCT) compound, cryo-sectioned at a thickness of 10 µm, and mounted using VECTASHIELD Antifade Medium containing DAPI. The samples were subsequently analyzed by fluorescence microscopy (LEICA SP8) to assess peptide penetration and distribution.

Optimized for binding to damaged collagen

The NeoHelix Regenerate peptide sequence was rationally optimized to achieve high-affinity binding to damaged collagen. An *in vitro* binding assay performed on denatured collagen demonstrated that 4 Gly-X-Y repeats (corresponding to the NeoHelix Regenerate peptide) are the optimal binding sequence.

Reducing the number of repeats to 3 or increasing it to 5 resulted in a decreased binding affinity (Fig. 6). In addition, the integrin-binding sequence alone (GFOGER) did not exhibit any detectable binding to collagen, highlighting the critical role of the optimized Gly-X-Y repeat architecture in collagen interaction.

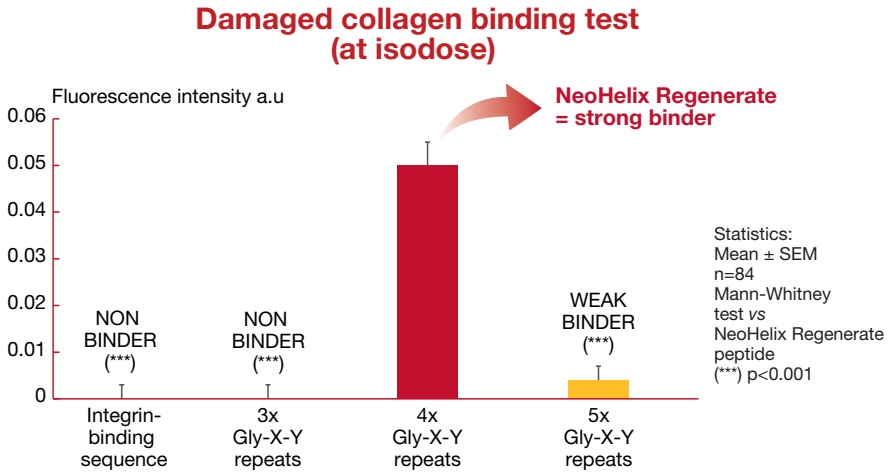


Figure 6 - *In vitro* binding assay to damaged collagen of different peptides, including peptides containing different numbers of Gly-X-Y repeats as well as the integrin-binding sequence alone.

MATERIALS&METHODS

Study number

EMB230801-2025-036-CC

Tested products

Fluorescein-tagged peptides containing different numbers of Gly-X-Y repeats (MOL_0001130_001, MOL_0001138_001, MOL_0001139_001) and integrin-binding site sequence (MOL_0001116_001) were tested diluted in PBS at the same concentration of 15 µM (isodose).

Binding test

The binding assay was performed using a 96-well plate coated with collagen type I (Nunc, Thermo Scientific). Collagen I immobilized on coated plates is partially denatured and not fully maintained in its native triple-helical conformation. Indeed, the coating process, including acid dissolution required for collagen solubilization, surface drying, and subsequent sterilization procedures such as UV exposure or rinsing with ethanol alter the native collagen structure. These commonly applied steps induce structurally altered collagen on the coated surface. 50 µL of each fluorescent peptide solution was added to each well (4 wells per condition), followed by overnight incubation at 32°C to allow peptide binding. The wells were subsequently washed three times with 100 µL of TBST2 (Tris-buffered saline containing 0.05% Tween 20 and 1 mM $MnCl_2$; Sigma). Fluorescence was then measured using a plate reader (excitation at 493 nm and emission at 517 nm), with 21 measurement points per well.

Results and Statistics

Results are expressed as Mean value ± SEM (n=84) The statistics were performed according to SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California, United States of America). Statistical analysis was assessed by Mann-Whitney test with threshold of significance set to 5% ($p<0.05$).

With selective binding to damaged collagen

A thermal shock experiment designed to denature collagen demonstrates the selective binding behavior of NeoHelix Regenerate (Fig. 7). Before heat exposure, the skin exhibited a high level of intact collagen I, specifically recognized by the antibody used in the experiment (red signal). Under these conditions, NeoHelix Regenerate (green signal) showed only minimal binding, reflecting the limited availability of binding sites on undamaged collagen. After the thermal shock, collagen I became significantly denatured, as evidenced by the diminished red signal, and numerous unfolded regions were exposed. Consequently, a marked increase in NeoHelix Regenerate binding was observed (green signal increase), directly linked to the newly accessible sites on damaged collagen fibers.

This experiment clearly highlights the strong selectivity of NeoHelix Regenerate: it targets, recognizes, and binds to damaged collagen rather than to its intact form.

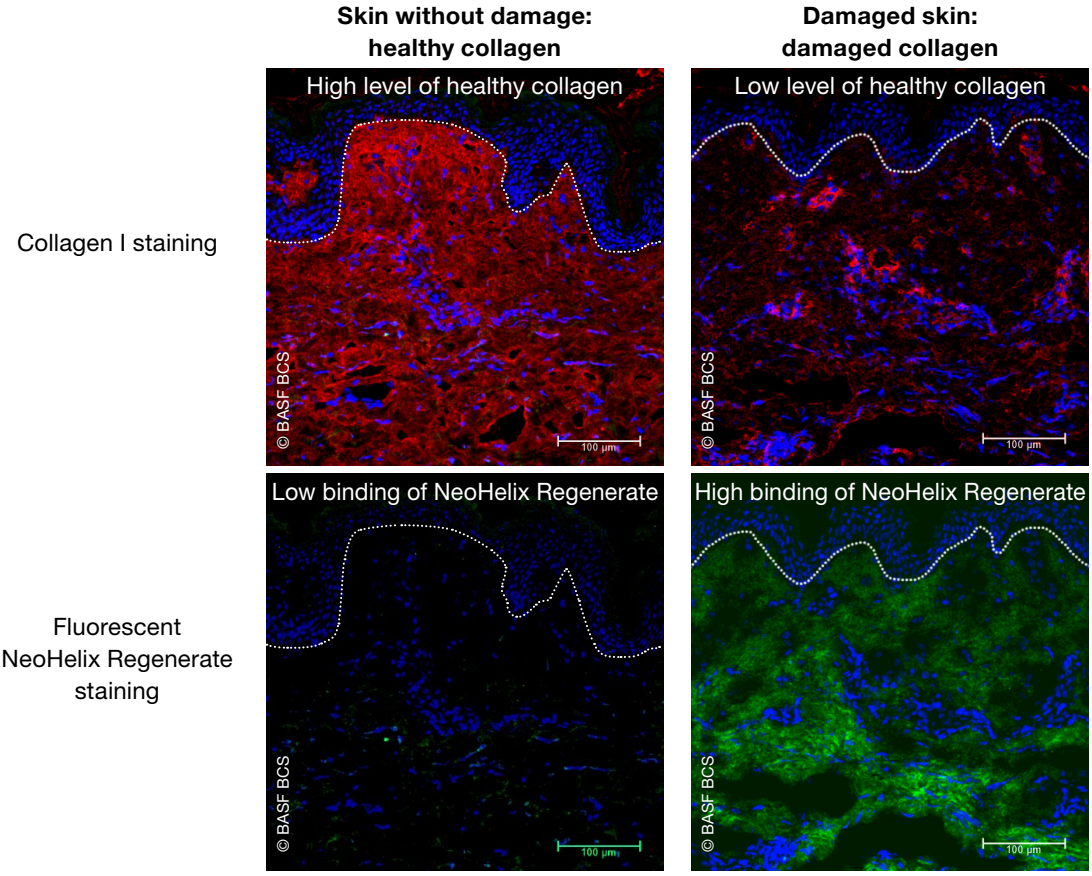


Figure 7 - Representative images illustrating the selective binding of NeoHelix Regenerate to damaged collagen (in green: fluorescein-tagged NeoHelix Regenerate peptide; in red: healthy collagen; in blue: DAPI-stained nuclei).

MATERIALS&METHODS

Study number
EMB230801-2025-065-IM

Tested products
Fluorescein-tagged NeoHelix Regenerate peptide (MOL-0001130-001) diluted in PBS at a concentration of 20 µM.

Treatment
Human skin samples were obtained from frozen tissue specimens. Samples were cryo-sectioned at a thickness of 10 µm using a cryostat, and sections were collected onto Superfrost Plus microscope slides. The slides were air-dried for 30 minutes at room temperature, then the sections were fixed in 4% paraformaldehyde (PFA) for 20 minutes at room temperature. Following fixation, sections were washed three times for 10 minutes in phosphate-buffered saline (PBS).

The slides were then heated in an oven at 80°C for 20 minutes (heat shock for denaturing collagen) and allowed to cool to room temperature. The sections were permeabilized with 0.3% Triton X-100 for 10 minutes and blocked in 2% serum for 1 hour at room temperature. They were incubated with an anti-collagen I primary antibody (Novotec) for 2 hours, followed by an Alexa Fluor 594-conjugated secondary antibody (InVitrogen) for 45 minutes, with PBS washes between steps.

The sections were then incubated overnight at 4°C with the Fluorescein-tagged NeoHelix Regenerate peptide at a final concentration of 20 µM. After incubation, slides were washed three times for 10 minutes in PBS and mounted using VECTASHIELD Antifade Mounting Medium containing DAPI for nucleus staining.

Visualization
Fluorescence imaging was performed using a Leica TCS SP8 confocal microscope equipped with a 20X objective.

Triggering signaling pathways through adhesion to collagen via integrin-mediated binding

Through talin expression and pro-collagen I synthesis assays, we evaluated the ability of NeoHelix Regenerate to effectively interact with integrins.

Talin is a cytoplasmic protein that links integrins to the actin cytoskeleton. When collagen binds to integrins, talin is recruited and activated, inducing conformational changes in integrins. This activation enhances integrin affinity for collagen and strengthens the signal transmitted between the extracellular matrix and integrins, playing a fundamental role in cellular signaling [Chen *et al*, 2019; Chakraborty *et al*, 2019].

We observed that following UVA irradiation, talin expression was reduced compared with the untreated, non-irradiated control (Fig. 8). UVA exposure is assumed to induce collagen denaturation, thereby weakening collagen-integrin interactions and impairing downstream integrin-mediated signaling. Upon treatment with NeoHelix Regenerate, talin synthesis is restored (+202%, SEM ± 12%, p<0.001) *vs* untreated control with UVA (normalized at 100%, SEM ± 3%), indicating a reactivation of integrin-associated signaling pathways.

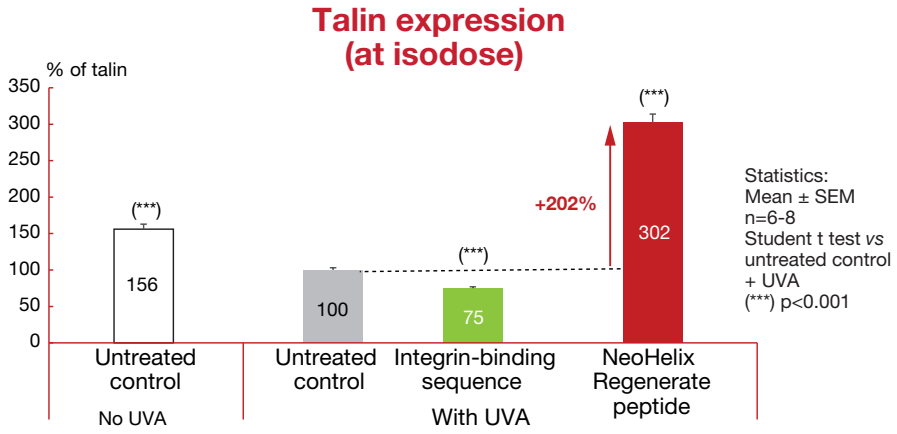


Figure 8 - Talin expression assay on human fibroblasts in an UVA stress model.

As expected, the peptide of the integrin-binding sequence alone had no effect on talin. Indeed, this peptide alone has not the correct three-dimensional form to ensure effective binding to integrins and functional effect [Knight *et al*, 2000; Emsley et al, 2004; Jokinen *et al*, 2010].

Similarly, on pro-collagen I synthesis by fibroblasts (known to be triggered via integrin-mediated pathways [Elango *et al*, 2022], only NeoHelix Regenerate demonstrated a significant boosting effect (+80%, SEM ± 12%, p<0.001, *vs* the untreated control normalized at 100% SEM ± 13%) whereas the peptide of the integrin-binding sequence alone had no effect (Fig. 9).

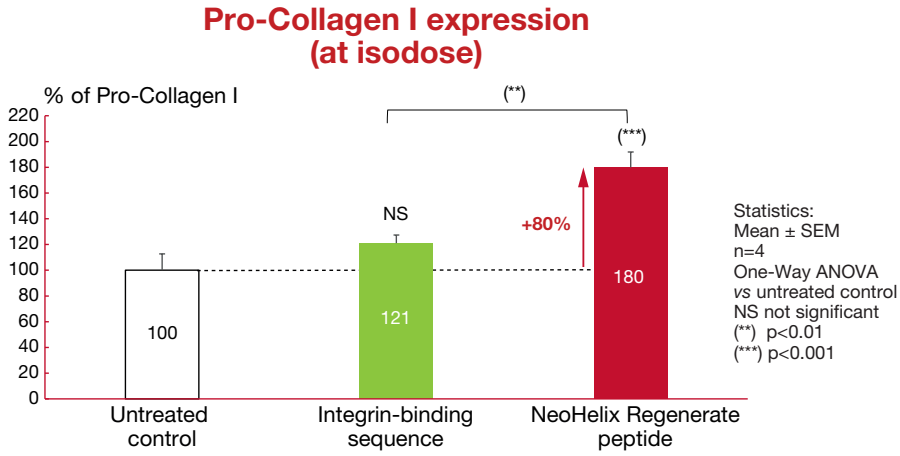


Figure 9 - Pro-collagen I expression assay on human fibroblasts.

Even indirectly, these results suggest that NeoHelix Regenerate, through its specific combination of collagen and integrin-binding motifs, enables three dimensional structure recognition and efficient binding to cellular integrin receptors.

This interaction facilitates fibroblast adhesion to collagen and triggers integrin-mediated signaling pathways, leading to downstream integrin-dependent cellular processes.

MATERIALS&METHODS

Study number

EMB230801-2025-005-PROT (Talin)
EMB230801-2025-016-PROT (Pro-Col I)

Talin assay

Tested product

NeoHelix Regenerate peptide (MOL_0001111_002) and integrin-binding site peptide (MOL_0000965_001) were tested diluted in PBS at a concentration of 0.4 µM.

Cell culture and treatment

Normal human fibroblasts derived from a 19-year-old female donor were cultured in 12-well plates and exposed or not to UVA irradiation (10 J/cm²). Cells were then treated for 6h with NeoHelix Regenerate peptide or with the integrin-binding peptide or left untreated (UVA control). Culture medium was subsequently replaced with fresh medium without active compounds. 48h after the start of the experiment, cells were harvested and lysed.

Quantification of talin by Western blot

Protein concentration was determined by bicinchoninic assay (BCA protein assay), then all the samples were adjusted to the same protein concentration and deposited at the same quantities in each capillary. Samples and reagents were prepared according to the recommended manufacturer's instructions. Talin expression was analyzed by a capillary electrophoresis-based protein analysis system (Sally Sue Instrument, ProteinSimple, San Jose, California, USA) using a specific anti-talin antibody and chemiluminescent detection. Protein levels were quantified using Compass software. Data represent six to eight independent replicates (n = 6-8).

Results and statistics

Talin expression level was expressed as mean percentage ± standard error on the mean (n=6-8) compared to the untreated control with UVA stress, The statistics were performed according to SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California, United States of America). Statistical analysis between conditions has been assessed by a Student t test with threshold of significance set to 5% ($p<0.05$)

Pro-Col I assay

Tested products

NeoHelix Regenerate peptide (MOL_0001111_002) and integrin-binding sequence peptide (MOL_0000965_001) were tested diluted in PBS at a concentration of 5 µM.

Cell culture and treatment

Normal human fibroblasts derived from a 19-year-old female donor were seeded in 96 well plates and cultured for 4 days at 37°C and 5% CO_2. Cells were then treated for 6 h with either NeoHelix Regenerate peptide (5 µM), integrin-binding sequence peptide (5 µM) or left untreated (untreated control). Culture medium was subsequently replaced with fresh medium without active compounds.

Quantification of pro-collagen I

After 48 h of incubation supernatants were collected to quantify Pro-Collagen I using a Human Pro-Collagen I ELISA kit (PICP, Abcam).
A cell lysis reagent was subsequently added to each well to measure DNA content. DNA levels were then quantified using Quant-iT PicoGreen reagents (Thermo Fisher Scientific).

Results and statistics

Fluorescence values were normalized to DNA content to obtain a stimulation ratio for each well. The percentage of stimulation of pro-Collagen I synthesis was calculated relative to the untreated control and expressed as mean percentage ± standard error on the mean (n=4). The statistics were performed according to SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California United States of America). Statistical analysis between conditions has been assessed by a Student t test with threshold of significance set to 5% ($p<0.05$).

NeoHelix Regenerate introduces a new class of cosmetic active ingredient inspired by the biomedical Collagen Hybridizing Peptide technology. It is a unique patented peptide with a dual sequence designed to target precisely damaged collagen and restore cell-collagen signaling pathways for collagen repair and skin regeneration.

Safety / tolerability of the product

NeoHelix Regenerate was evaluated to ensure its safety in the recommended conditions of use.

Description

NeoHelix Regenerate is a preservative free powder containing ~200 ppm of the synthetic biomimetic Collagen Hybridizing Peptide designed for collagen repair and skin regeneration, formulated with mannitol as an excipient.

INCI name: Mannitol (and) Acetyl Tetra (Tripeptide-29) Dipeptide-73 Hydroxyproline Tripeptide-122 Tetra (Tripeptide-29)

Naturalness content (according to ISO 16128): 99.9% from natural origin

Recommended dose of use: 1%



Example: illustration of a commercial sample of NeoHelix Regenerate incorporated into a face cream and a toning lotion at 1%.



Demonstrated efficacy

NeoHelix Regenerate supports skin regeneration by modulating key biological processes across the epidermis, dermis, and hypodermis, as demonstrated *in vitro*. Its efficacy was further validated *in vivo* through two clinical studies: one confirming its skin regeneration activity and the other demonstrating its ability to improve visible signs of skin aging.

In vitro the properties of NeoHelix Regenerate were demonstrated on:

- the acceleration on epidermal regeneration in human skin explants,
- the reduction of the pro-inflammatory cytokine IL-8 release by human keratinocytes,
- the stimulation of collagen I and III synthesis in the ECM on 3D-reconstructed dermis,
- the protection of the proteins of the matrix from glycation in human skin explants,
- the reactivation of cellular communication through the stimulation of leptin in 3D-vascularized tissue organoid model.

In vivo, in a placebo-controlled study through skin biopsy analyses, we demonstrated NeoHelix Regenerate's ability to promote skin regeneration and support collagen repair. A second placebo-controlled study, including the comparison with a leading peptide blend benchmark, highlighted the properties of NeoHelix Regenerate on wrinkles reduction and improvement of skin tonicity and thickness.

EFFICACY

Epidermal regeneration booster

ex vivo

OBJECTIVE

A compromised extracellular matrix, particularly when the collagen network is damaged, slows epidermal renewal, leading to an epidermis with reduced repair capacity [Huang *et al*, 2022; Potekaev *et al*, 2021; Watt *et al*, 2011]. Building on this understanding, wound-healing models based on biopsies offer a relevant approach to assess the regenerative efficacy of active ingredients [Choudhary *et al*, 2024], as the growth of the neo-epidermis provides a direct indicator of tissue repair [Wang *et al*, 2019].

In parallel, K17 (Keratin 17) serves as a highly specific injury-induced marker, as its expression occurs exclusively following skin injury [Zhang *et al*, 2019]. Tracking K17 levels further characterizes the regenerative activity of an ingredient and its impact on epithelial recovery.

In this context, the aim of the present study was to evaluate NeoHelix Regenerate's ability to enhance both neo-epidermis growth and K17 expression in a human skin explant wound-healing model.

RESULTS & DISCUSSION

Neo-epidermis length measurement

Statistical analysis revealed a significantly greater increase in neo-epidermis length when treated topically with NeoHelix Regenerate (Fig. 10 and 11). 5 days post-lesion, NeoHelix Regenerate at 1% resulted in a 35% increase compared to the untreated control (322 µm ± SD 126 µm *vs* 239 µm ± SD 73 µm, p<0.1). After 8 days, the increase reached 43% (305 µm ± SD 114 µm *vs* 214 µm ± SD 47 µm, p<0.05) *vs* the untreated control.

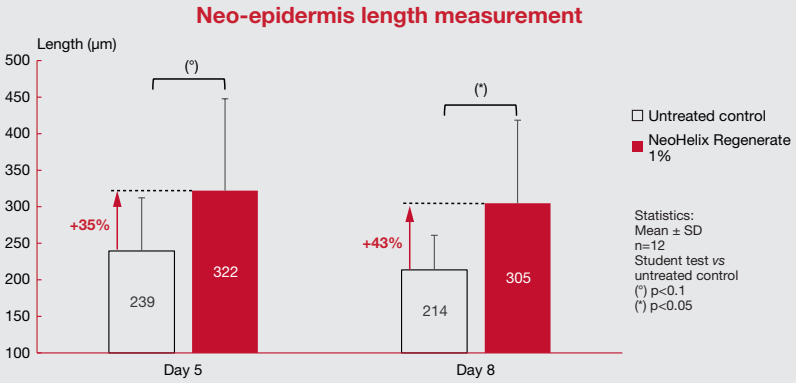


Figure 10 - Neo-epidermis length measurement in wounded skin biopsies.

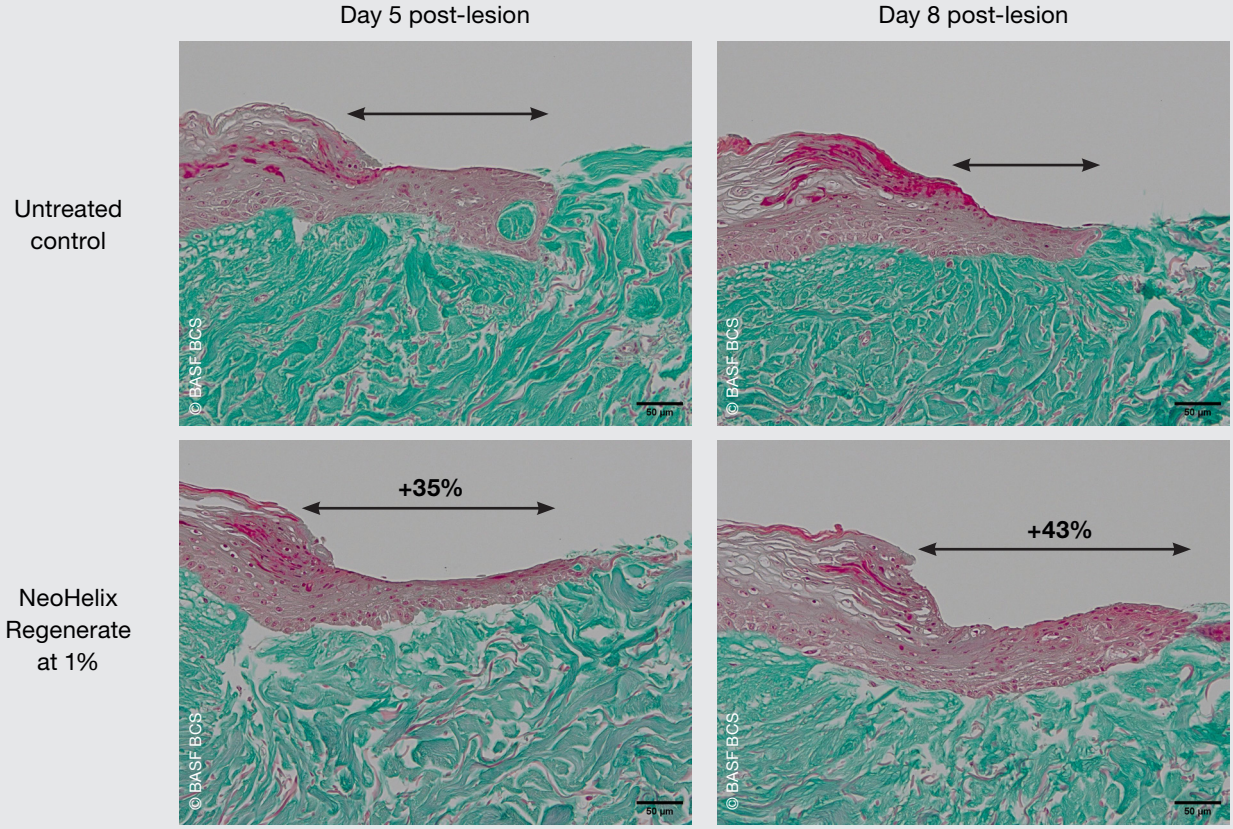


Figure 11 - Illustrative pictures of neo-epidermis length progression.

ex vivo

RESULTS & DISCUSSION

Stimulation of K17 in the neo-epidermis

Statistical analysis indicated a significantly greater increase in K17 expression when treated topically with NeoHelix Regenerate (Fig. 12 and 13). 5 days post-lesion, treatment with 1% NeoHelix Regenerate resulted in a +63% increase compared to the untreated control (SEM 7% *vs* 100% SEM ± 4%, p<0.01). After 8 days, the increase reached +71% *vs* the untreated control (SEM ± 15% *vs* 100% SEM ± 16%, p<0.001).

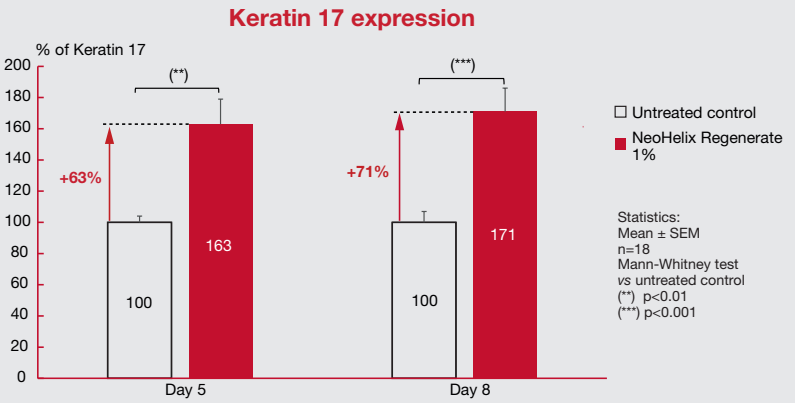


Figure 12 - Keratin 17 measurement in skin biopsies.

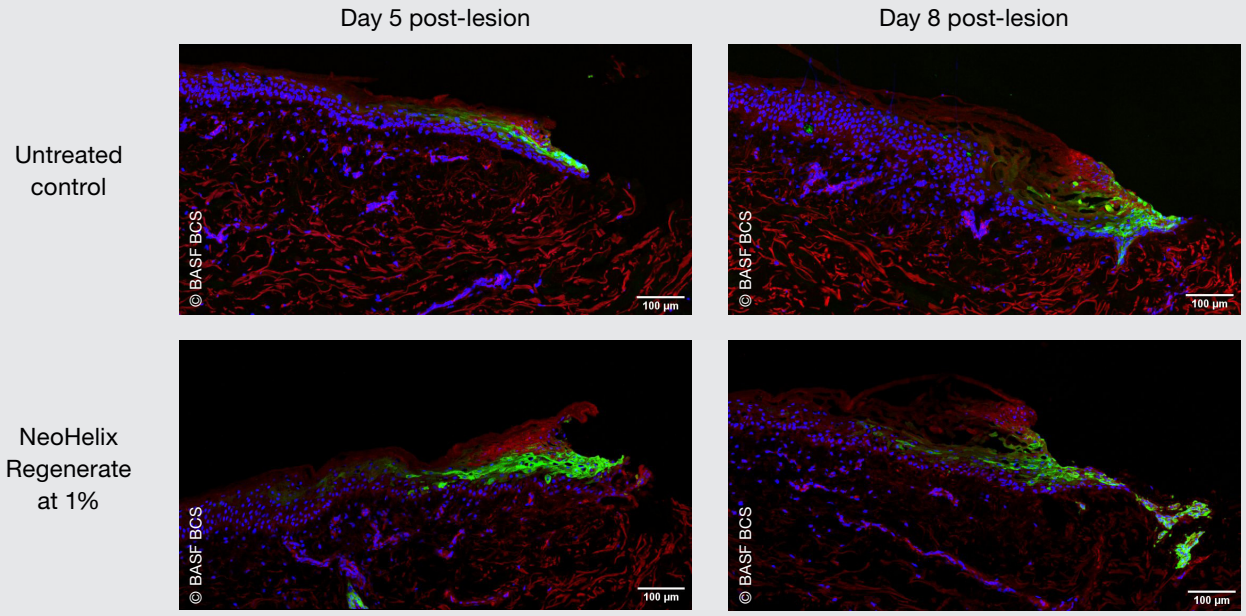


Figure 13 - Illustrative pictures of K17 immunostaining (in green: K17 staining; in red: Evans Blue was used as a contrast agent; in blue: DAPI-stained nuclei).

CONCLUSION

Ex vivo, in a wound-healing model, NeoHelix Regenerate at 1% stimulated neo-epidermis growth and increased K17 expression, both key indicators of active epidermal tissue repair. These results suggest that NeoHelix Regenerate helps restore epidermal regeneration processes, supporting a more resilient skin.

MATERIALS&METHODS

Study number

EMB230801-2024-123-HI

Tested products

NeoHelix Regenerate peptide (MOL_0001111_003) at 0.8 µM (corresponding to 1% of NeoHelix Regenerate BC10226) diluted in Dulbecco's phosphate-buffered saline solution (DPBS).

Cell culture

Rectangular explants of 10 x 15 mm (1.5 cm^2) were prepared from abdominoplasty tissue obtained from a 49-year-old woman. The explants were maintained in an appropriate culture medium for explant survival at 37°C in a humid atmosphere with 5% CO_2.

At Day 0 (start), two lesions per explant were created using a 3.5 mm biopsy punch at a depth sufficient to remove the epidermis and the superficial part of the dermis. Lesions were made immediately before product application.

At Day 0 (after lesion), Day 2, Day 5, and Day 7, the NeoHelix Regenerate peptide solution was applied topically at 3 µL per "wounded" explant of 1.5 cm^2 (≈ 2 mg/cm^2) and spread with a small spatula (n = 3 explants per condition). The culture medium was half-renewed on days 2, 5, and 7.

Injured explants without peptide solution application, referred to as untreated controls, served as controls for analysis on days 5 and 8.

On days 0, 5, and 8, explants were collected and cut into two parts. Half of the specimen was fixed in buffered formalin solution, while the remaining portion was embedded in OCT (Optimal Cutting Temperature) compound, frozen at -80°C for histological examination. Following a 24-hour period, the formalin-fixed samples were dehydrated and embedded in paraffin

Neo-epidermis length measurement method

The length of the synthesized neo-epidermis was assessed by examining 5 µm paraffin skin sections under a microscope after Masson's trichrome staining, using a Leica DMLB microscope equipped with an Olympus DP digitization device.

Keratin 17 staining method

10 µm cryosections were fixed in cold acetone at -20°C for 10 minutes to preserve tissue morphology. Following fixation, slides were rinsed three times for 10 minutes each in PBS to remove residual fixative. Then sections were incubated in blocking solution of 2% sheep serum for 2 hours at room temperature. Samples were then incubated with a primary antibody against keratin 17 for 1 hour at room temperature. After primary antibody incubation, sections were washed three times in PBS (10 minutes each) and incubated with an Alexa Fluor 488-conjugated secondary antibody for 45 minutes at room temperature. Slides were again rinsed three times in PBS to remove unbound secondary antibody. To visualize tissue architecture, sections were counterstained with Evan's Blue for 10 minutes, followed by PBS washes. Slides were mounted using Vectashield mounting medium containing DAPI to label nuclei.

Fluorescence imaging was performed using a Leica TCS SP8 confocal microscope coupled with an Olympus DP digitization device.

Results and statistics

The length of neo-epidermis was measured in micrometers (µm) utilizing ImageJ software. Data were expressed as length in µm and percentage changes relative to untreated control at 5- and 8-day post-lesion (D5 and D8) are calculated.

K17 levels were measured based on their staining intensity in the neo-epidermis region. The results are presented as the percentage change in K17 intensity compared to untreated control within the neo-epidermis area.

The results are presented as mean ± SD (standard deviation) for the neo-epidermis length and mean ± SEM (Standard Error of the Mean) for K17 quantification.

Statistical analysis of data was performed using SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California United States of America), with n = 12 images (3 explants per condition; 4 images per explant) for neo-epidermis length measurement and n=18 images (3 explants per condition; 5-7 images per explant) for K17 quantification. Statistical analysis between conditions has been assessed by a Student t test or Mann-Whitney test according Sigmaplot recommendation, with threshold of significance set to 5% ($p<0.05$).

EFFICACY

Decreased release of pro-inflammatory cytokines by keratinocytes

in vitro

OBJECTIVE

Inflammatory responses play a pivotal role in wound-healing and skin regeneration dynamics: although an early pro-inflammatory response is essential to clear debris and initiate tissue repair, excessive or prolonged cytokines expression can disturb this balance, delaying re-epithelialization and impairing optimal regeneration [Gillitzer *et al*, 2001; Ridiandries *et al*, 2018; Fernandez-Guarino *et al*, 2024]. As one of the key cytokines produced during cutaneous injury, Interleukin-8 (IL-8) drives neutrophil recruitment and amplifies the inflammatory cascade, and its sustained elevation is associated with chronic inflammatory disorders and compromised skin regeneration [Bickel,1993; Rembe *et al*, 2025].
Therefore, the objective of this study was to evaluate the effect of NeoHelix Regenerate on the release of IL-8 on stimulated keratinocytes.

RESULTS & DISCUSSION

The release by human keratinocytes of IL-8 was stimulated by a pro-inflammatory cocktail based on TNFα, IFNγ and Poly I:C (Fig. 14). Dexamethasone (100 µM) used as positive control, significantly reduced the release of the pro-inflammatory mediator IL-8 by keratinocytes and thus validated the assay (data not shown).
NeoHelix Regenerate at 0.125% had no cytotoxic effect on keratinocytes (MTT assay, data not shown). The treatment of human keratinocytes by NeoHelix Regenerate at 0.125% reduced the release of IL-8 induced by the pro-inflammatory cocktail by -31% (SD ± 8%, p<0.01) compared to the stressed-untreated control normalized at 100% (SD ± 12%) (Fig. 14).

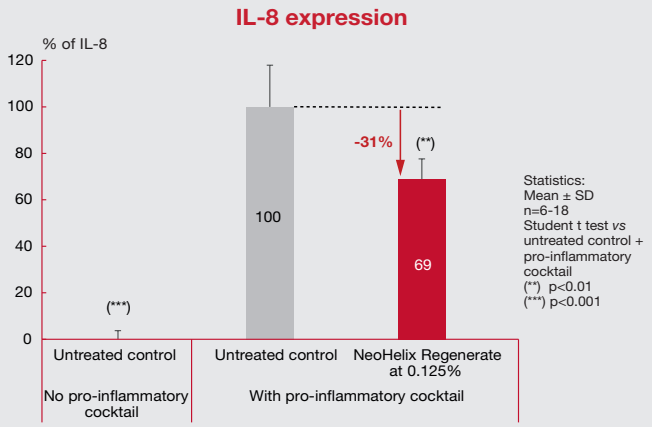


Figure 14 - IL-8 expression stimulated by a pro-inflammatory cocktail in human skin keratinocytes.

CONCLUSION

NeoHelix Regenerate has shown *in vitro* its ability to reduce the release of a pro-inflammatory mediator (IL-8) by stimulated keratinocytes, and thus its potential to protect and soothe the skin while supporting the natural processes of skin regeneration.

MATERIALS&METHODS

Study number
EMB230801-2024-047-IDCD

Tested products
NeoHelix Regenerate peptide (MOL_0001111_002) at 0.1 µM (corresponding to 0.125% of NeoHelix Regenerate BC10226) diluted in a standard nutrient-deprived culture medium.

Cell culture and treatment
Normal human keratinocytes from the abdominal area of a 64-year-old woman were seeded into 96-well plates and incubated at 37°C and 5% CO_2 for 3 days. The cells were first incubated, or not, for 24 hours with the NeoHelix Regenerate peptide, and then for an additional 24 hours with a pro-inflammatory cocktail composed of TNFα, IFNγ, and Poly I:C, at concentrations sufficient to significantly stimulate IL-8 release by keratinocytes. After the two 24-hour incubation phases, the supernatant was collected and analyzed using Revvity's AlphaLISA IL-8 technology.

Results and statistics
The percentage of IL-8 inhibition is determined by first subtracting the basal IL-8 concentration and then normalizing the IL-8 concentration to the pro-inflammatory cocktail untreated control (stressed untreated control) set at 100%.
The results are presented as mean ± SD (standard deviation). Statistical analysis of data was performed using SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California United States of America), with n = 6-18. Statistical analysis between conditions has been assessed by a Student t test with threshold of significance set to 5% (p<0.05).

EFFICACY

Boosted collagen production by fibroblasts

in vitro

OBJECTIVE

During intrinsic and extrinsic aging, collagen within the ECM progressively deteriorates mainly by oxidation, glycation, UV exposure and pollution.
It undergoes loss, damage, and structural disorganization, leading to weakened skin architecture, reduced density, and wrinkle formation [Fisher *et al*, 2023; Martic *et al*, 2022; Mahmoud *et al*, 2021; Tu *et al*, 2016; Quan *et al*, 2015].
This degradation results primarily from the damages inflicted on ECM fibers, especially collagen fibers, as their synthesis declines while their breakdown by matrix metalloproteases increases.
The aim of this study was to assess NeoHelix Regenerate's capacity to enhance both the synthesis of collagen I and III by fibroblasts in a 3D-reconstructed dermis.

RESULTS & DISCUSSION

NeoHelix Regenerate at 0.5% significantly stimulated collagen production by human fibroblasts in a 3D-dermis model (Fig. 15 and 16). Compared to the untreated control, NeoHelix Regenerate increased collagen I synthesis by +119% (SEM ± 30%, p<0.001) and collagen III synthesis by +240% (SEM ± 64%, p<0.001) *vs* untreated control normalized at 100% (SEM ± 4%).

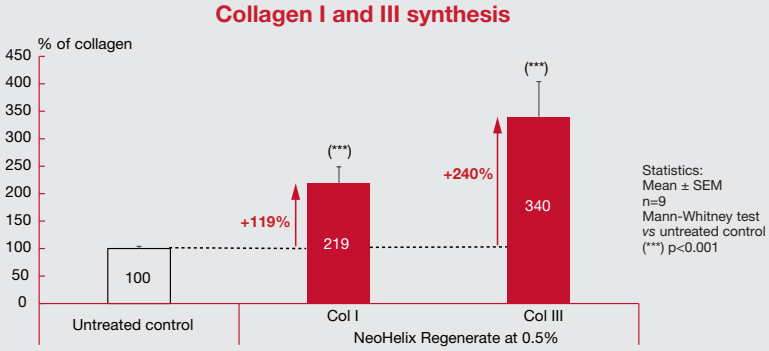


Figure 15 - Collagen I and III synthesis in a reconstructed dermis model.

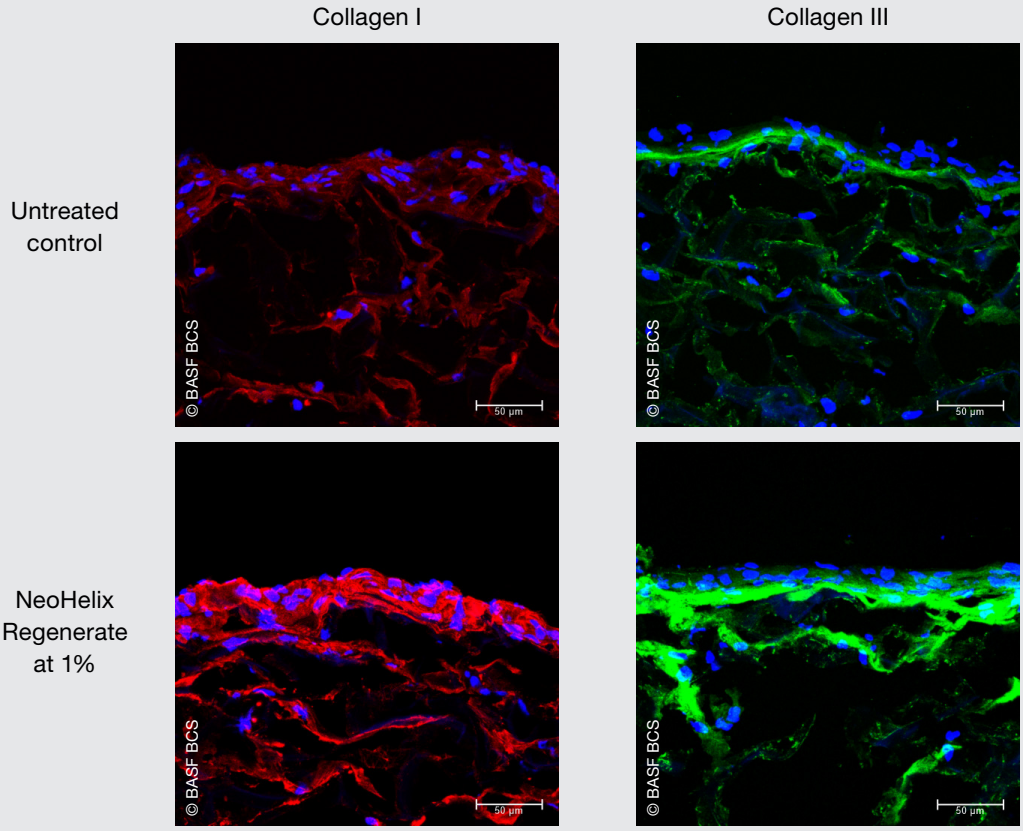


Figure 16 - Illustrative pictures of Collagen I and collagen III immunostaining in a reconstructed dermis model at D28 (in red: collagen I staining, in green: collagen III staining; in blue: DAPI stained nuclei).

CONCLUSION

NeoHelix Regenerate demonstrated a strong ability to boost the synthesis of key structural collagens in the extracellular matrix, which are essential for maintaining skin density and firmness, and overall skin regeneration capacity.

MATERIALS&METHODS

Study number
EMB230801-2024-078-MEXP

Tested products
NeoHelix Regenerate peptide (MOL_0001111_003) at 0.4 µM (corresponding to 0.5% of NeoHelix Regenerate BC10226) diluted in PBS.

3D dermis culture and treatment
Human reconstructed dermis models (Mimederm) were obtained by seeding fibroblasts isolated from a female donor (18 years old) onto an acellular matrix composed of collagen, chitosan, and glucosaminoglycan (Mimedisc). The constructs were cultured in a specific cell growth medium consisting of DMEM (Dulbecco's Modified Eagle Medium)/F12 supplemented with 10% FBS and antibiotics. Two conditions were tested (each n=3): the untreated control and the NeoHelix Regenerate peptide treatment applied every two days from day 7 to day 28. The experiment stopped on day 28.

Collagen I and collagen III quantification
The expression of collagen I and III was evaluated by immunostaining followed by image based quantification. Tissue cryosections (10 µm) were fixed in acetone at -20°C for 10 minutes, then incubated with 2% serum for 2 hours at room temperature. After washing, the sections were incubated for 2 hours at room temperature with anti collagen I antibodies (Invitrogen) and anti collagen III antibodies (Novotec). Secondary antibodies (ThermoFisher) were then applied. Finally, the slides were mounted with an antifade medium (Vectashield) containing DAPI (4',6-diamidino-2-phenylindole) for nuclear staining and analyzed by fluorescence microscopy (LEICA SP8).

Results and statistics
The expression of collagen I and collagen III was quantified by image analysis using FIJI software (ImageJ2, open source). Results are presented as mean ± SEM (standard error of the mean). A total of 9 images were acquired per condition (3 reconstructed dermis per condition and 3 images per reconstructed dermis).
Statistical analysis was performed using SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California United States of America). A two-tailed test (Mann-Whitney) was used to compare treated samples with untreated control according to software recommendation. The significance threshold was set at 5% ($p<0.05$).

EFFICACY

Matrix protection from glycation

ex vivo

OBJECTIVE

Glycation is a slow and irreversible process in which circulating sugars bind to collagen fibers, forming Advanced Glycation End-products (AGEs) that stiffen the matrix and hinder its degradation, leading to their progressive accumulation [Danby, 2010]. These crosslinked fibers disrupt dermal homeostasis as AGEs also exacerbate skin aging by stimulating MMPs and increasing oxidative stress, further accelerating collagen breakdown [Han *et al*, 2016; Zheng *et al*, 2022; Kasper *et al*, 2001].
In this context, the aim of the present study was to evaluate NeoHelix Regenerate's ability to protect the skin from ribose-induced glycation in a human skin explant model. Briefly, human skin explant was co-incubated with ribose and NeoHelix Regenerate, to evaluate its protective effect against glycation reaction.

RESULTS & DISCUSSION

This glycation assay was validated by two positive controls, rutin at 1 mM and aminoguanidine at 10 mM, which both significantly reduced protein glycation under the same experimental conditions (data not shown). Compared with the glycated control normalized at 100% (SD ± 1%), NeoHelix Regenerate reduced protein glycation by 66% at 0.1% (SD ± 4%, p<0.001) and by 73% at 0.5% (SD ± 14%, p<0.001).

CONCLUSION

NeoHelix Regenerate helps preserve the extracellular matrix by preventing glycation-induced damage, thereby maintaining its structural integrity and overall quality.

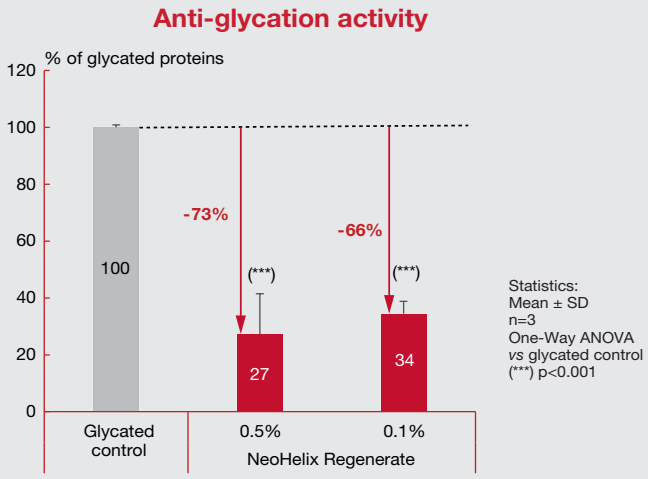


Figure 17 - Protective effect against protein glycation of a skin explant.

MATERIALS&METHODS

Study number
EMB230801-2024-134-CC

Tested products
NeoHelix Regenerate peptide (MOL_0001111_003) at 0.4 and 0.08 µM (corresponding to 0.5% and 0.1% of NeoHelix Regenerate BC10226) diluted in PBS.

Ribose-Induced Glycation Assay and quantification of glycated proteins
Whole human skin explants (5 mm diameter) from a 43-year-old donor were used for glycation assays. After 4 freeze–thaw cycles to remove residual enzymatic activity, explants were incubated for 14 days at 45°C with or without 60 mM ribose, and in the absence (Glycated control) or presence of NeoHelix Regenerate peptide. Three explants were used per condition.
At the end of incubation, glycated proteins were quantified using a colorimetric method [Kobayashi *et al*, 1996].

Results and statistics
Results are calculated in percentage of glycated protein *vs* glycated protein control (100%) and are represented as mean ± SD (standard deviation) with n=3. Statistical analysis was performed using SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California United States of America) using a One-Way ANOVA followed by a Holm Sidak test according to software recommendation. The significance threshold was set at 5% (p<0.05).

EFFICACY

Boosted leptin in adipocytes

ex vivo

OBJECTIVE

Adipocytes regulate fibroblast activity through adipokines such as leptin and adiponectin [Li *et al*, 2005]. More specifically, leptin secreted by adipocytes activates dermal fibroblasts, enhancing the production of collagen and hyaluronic acid [Ezure *et al*, 2007]. Maintaining sufficient leptin levels becomes particularly critical when collagen is damaged, as leptin helps reactivate fibroblasts, supports collagen and hyaluronic acid synthesis under stress conditions, and restores deep skin communication essential for effective tissue regeneration.
In this context, the aim of the present study was to evaluate NeoHelix Regenerate's ability to restore basal leptin gene expression levels in a 3D-vascularized tissue organoid model stressed with TNFα.

RESULTS & DISCUSSION

TNFα, a potent pro inflammatory cytokine, markedly suppresses leptin secretion, as reflected by the sharp decline observed in the TNFα-treated control group (20% SEM ± 3% *vs* 100% SEM ± 49% for the untreated control, p<0.1). The assay was validated using dexamethasone 0.25 µM as a positive control, confirming the expected inflammatory pathway (data not shown). NeoHelix Regenerate counteracts the TNFα induced suppression by restoring leptin secretion to the basal level (117% SEM ± 32%). This 5.8 fold increase (p<0.05) *vs* the stressed control with TNFα-demonstrates NeoHelix Regenerate ability to reactivate adipocyte signaling despite inflammatory stress, thereby establishing a physiological leptin environment essential for dermal communication and regeneration.

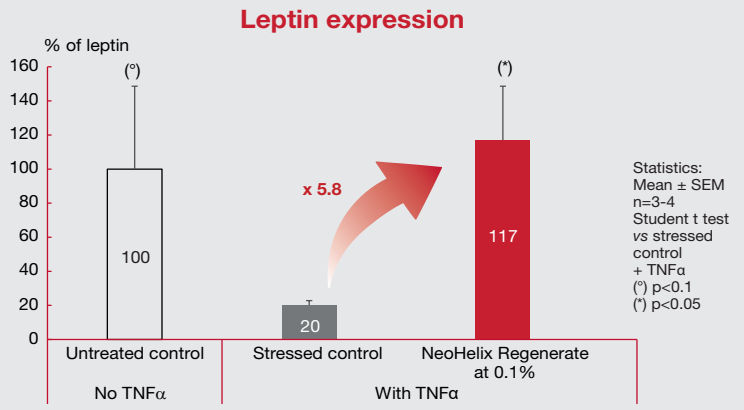


Figure 18 - Leptin gene expression in a 3D-stressed-vascularized tissue organoid model.

CONCLUSION

NeoHelix Regenerate demonstrated a strong ability to maintain sufficient leptin levels under stress conditions, helping to reactivate fibroblasts and support collagen and hyaluronic acid synthesis essential for skin regeneration.

MATERIALS&METHODS

Study number
EMB230801-2024-139-CC

Tested products
NeoHelix Regenerate peptide (MOL_0001111_003) at 0.08 µM (corresponding to 0.1% of NeoHelix Regenerate BC10226) diluted in water.

3D-vascularized tissue organoid model culture and treatment
Human subcutaneous vascularized adipose tissue models were generated from fresh lipoaspirate material obtained from a donor (female, 65 years old). Centrifugation was then performed to separate the adipose tissue and cell fractions. The cellular fraction, containing a highly proliferative stem cell population, was purified, and the adipose tissue fraction was mechanically dissociated using a medical device that reduces tissue size to obtain lobular structures. Finally, the cell and tissue fractions (adipocyte preparation) were mixed and incubated at 37°C, 5% CO_2 in an optimized medium (EGM Plus, Lonza) for 2 weeks prior to treatment.
NeoHelix Regenerate (0.08 µM) or dexamethasone (0.25 µM) is added at the tested concentration in a basal medium (EGM Endothelial Cell Growth Medium) supplemented with FBS, heparin, and ascorbic acid, with or without the addition of TNFα at a dose sufficient to induce inflammatory stress characterized by a significant decrease in leptin gene expression.
On Day 0, 50 mg of adipocyte preparation were encapsulated in low melting temperature agarose and deposited into each well of a 24-well plate; 1 mL of basal medium containing or not TNFα and tested product was added. The medium (containing the test products) was renewed every 2-3 days for 10 days. The number of replicates per condition was set at n=3-4.

Leptin gene expression quantification
On Day 10, cell viability was assessed using a XTT assay kit (Roche), following the manufacturer's instructions (data not shown) and total RNA was extracted from n = 3 or 4 replicates per group using a RNA extraction kit (Zymoclean Gel RNA Recovery Kit, Ozyme), following the manufacturer's instructions. An RT qPCR assay (reverse transcription-polymerase chain reaction analysis and real-time PCR) was performed on each replicate in duplicate (n = 8 values per condition). Leptin gene expression was analyzed by real-time PCR using the Applied Biosystems StepOne Plus system for all replicates. The primer sequences used for leptin detection were as follows:
hLeptin Fw: 5'-AGGGAGACCGAGCGCTTTC-3'
hLeptin Rv: 5'-TGCATCTCCACACACCAAACC-3'

Results and statistics
Results are calculated as a percentage relative to the untreated control set at 100% and are expressed as mean ± SEM (standard error of the mean) with n=3-4. Statistical analysis of the data was performed using SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California United States of America). A Student's test was used according to software recommendation. The significance threshold was set at 5% (p<0.05).

EFFICACY

Acting on dermal renewal: repairing collagen, boosting hyaluronic acid

in vivo

OBJECTIVE

Collagen fibers are a central structural component of the dermal extracellular matrix, and their degradation is a defining feature of skin aging [Tobin, 2017; Shao *et al*, 2019]. External stresses such as UV radiation and pollution, along with internal drivers like oxidative stress and glycation, progressively damage collagen and weaken the overall matrix architecture [Fisher *et al*, 2023; Martic *et al*, 2022; Mahmoud *et al*, 2021; Tu *et al*, 2016; Quan *et al*, 2015]. This structural damage also disrupts fibroblast signaling: fibroblasts receive weakened mechanical and biochemical cues through integrins, reducing their ability to regenerate the extracellular matrix and decreasing the synthesis of key ECM components. A self amplifying cycle then ensues, accelerating enzymatic degradation, further compromising dermal integrity, and ultimately diminishing the skin's regenerative capacity [Vanari *et al*, 2006; Tu *et al*, 2016; Fisher *et al*, 2023].

Randomized twice daily self application on the inner forearm for 56 days.
Histological quantification of damaged collagens and hyaluronic acid content on punch biopsies by image analysis

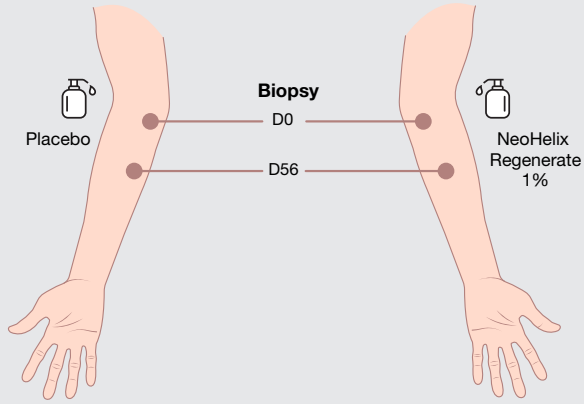


Figure 19 - Clinical study design.

The aim of the study was to evaluate, *in vivo*, the improvement of the extracellular matrix on healthy female volunteers (Caucasian origin), aged 60-70 years old. A double-blind, placebo-controlled clinical study was conducted on volunteers to evaluate the efficacy of NeoHelix Regenerate at 1%, based on skin biopsies taken from their forearms before and after treatment for 56 days (Fig. 19). The improvement of the extracellular matrix was quantified by the measurement of damaged collagens and hyaluronic acid contents on these biopsies.

RESULTS & DISCUSSION

The statistical analysis revealed a significant and potent reduction of the level of damaged collagen. After 56 days of application, NeoHelix Regenerate at 1% demonstrated a -41% reduction ($p<0.01$) in damaged collagens compared to baseline, and a -25% reduction ($p<0.01$) compared to the placebo formulation (Fig. 20 and 21).

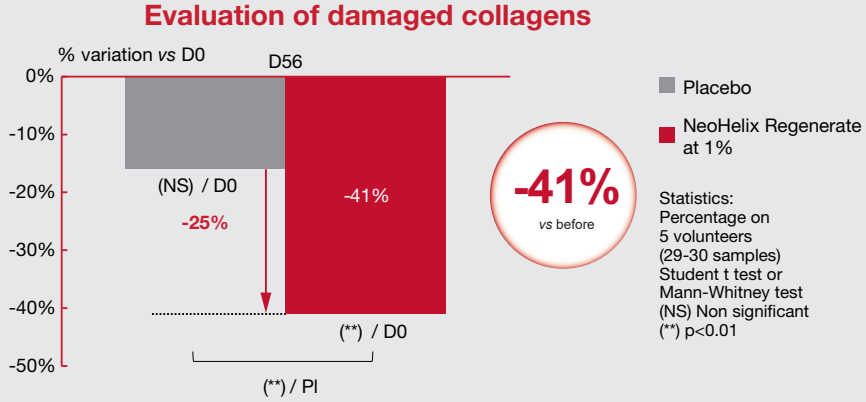


Figure 20 - Clinical study: decrease of the level of damaged collagen after 56 days of treatment with NeoHelix Regenerate at 1% or Placebo. Application on the inner forearm. Quantification using a fluorescent probe (F-CHP) specific to damaged collagen on skin biopsies and image analysis.

in vivo

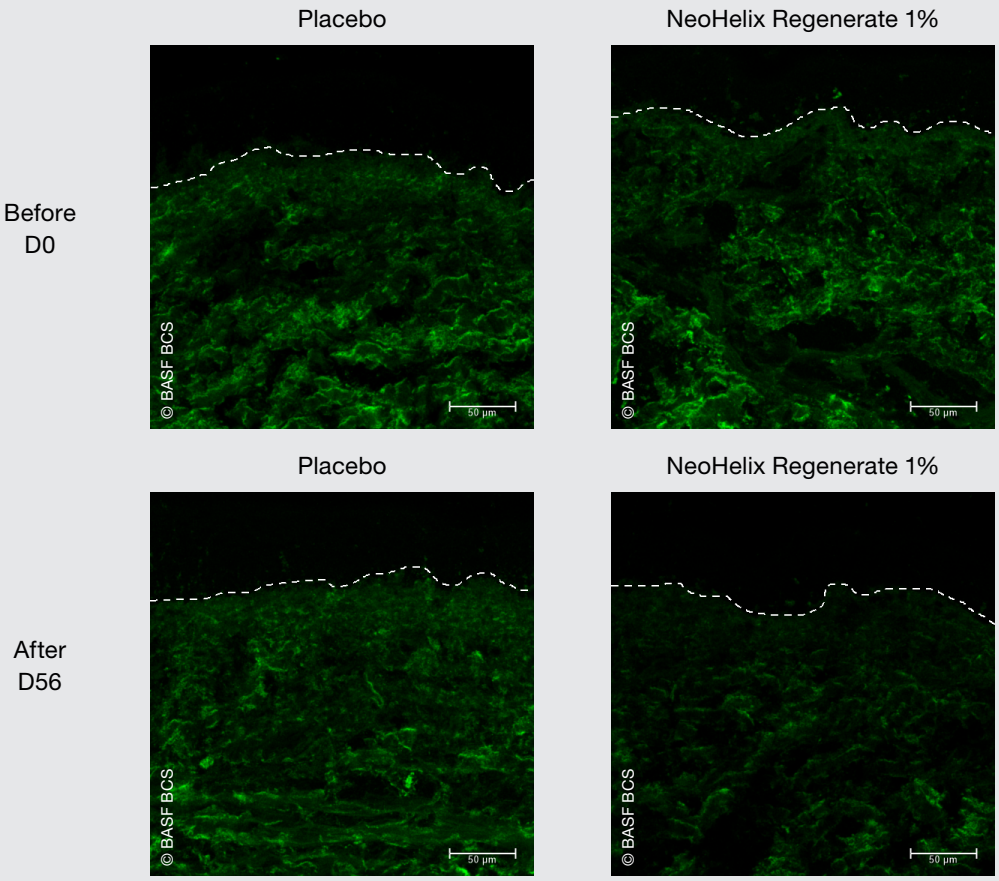


Figure 21 - Illustrative pictures of damaged collagens staining of skin biopsies at D0 and D56, volunteer number 3 (in green: damaged collagen staining; the epidermal-dermal junction is represented by a dotted line).

After 56 days of treatment, both biopsies showed also a strong increase in hyaluronic acid levels in the dermis. The treatment with NeoHelix Regenerate at 1% resulted in a statistically significant +65% ($p<0.001$) increase compared to baseline, and in a +35% ($p<0.05$) greater improvement compared to the placebo formulation (Fig. 22 and 23).

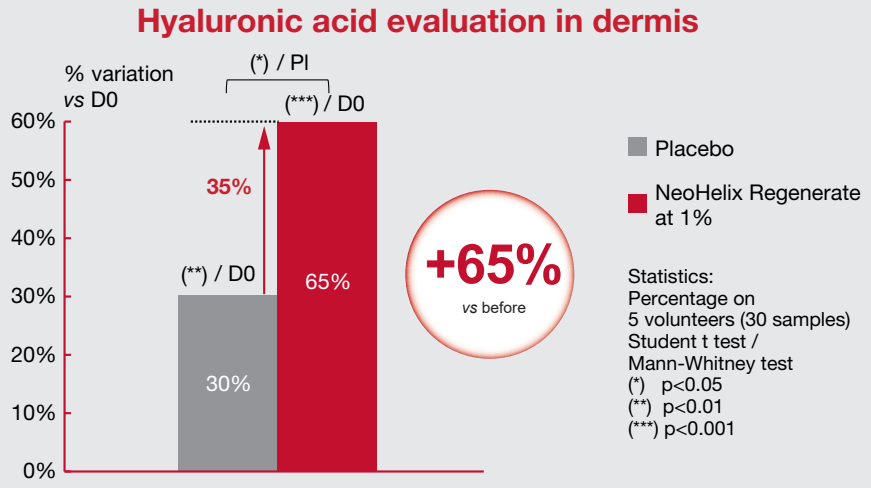


Figure 22 - Clinical study: increase of the level of hyaluronic acid after 56 days of treatment with NeoHelix Regenerate at 1% or Placebo. Application on the inner forearm. Quantification using a biotinylated probe specific to hyaluronic acid on skin biopsies and image analysis.

in vivo

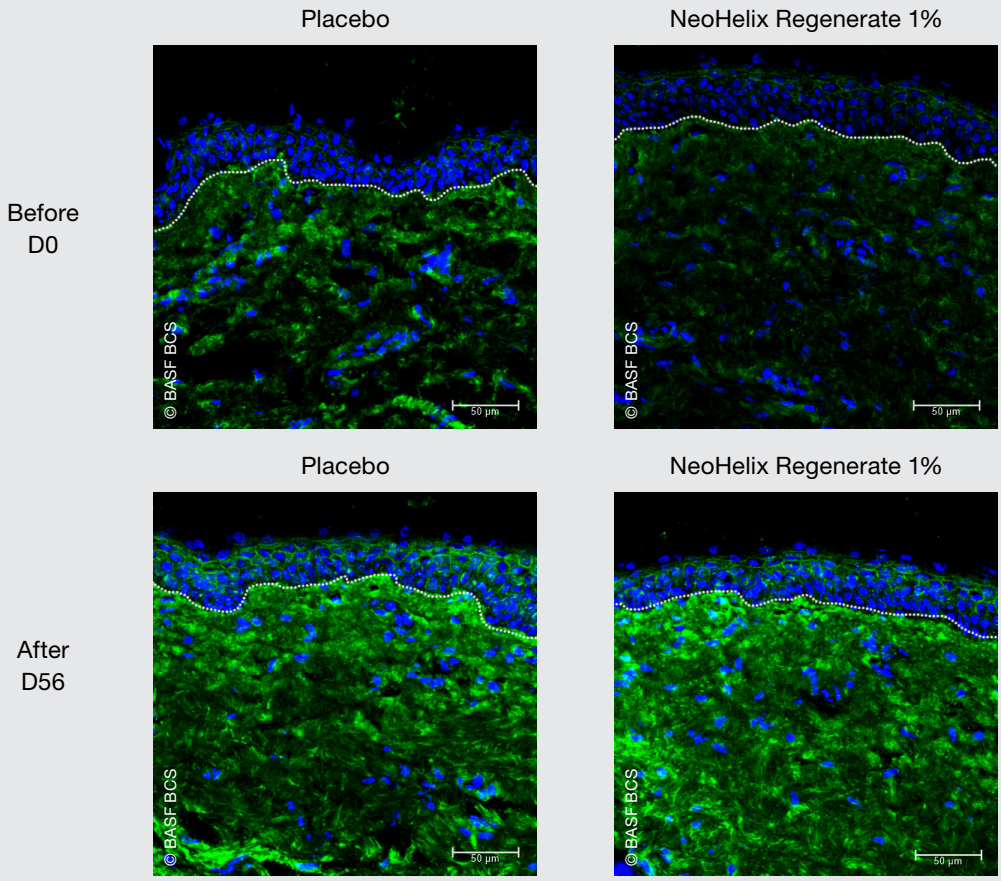


Figure 23 - Illustrative pictures of hyaluronic acid staining of skin biopsies at D0 and D56, volunteer number 4 (in green: hyaluronic acid staining; in blue: DAPI stained nuclei; the epidermal-dermal junction is represented by a dotted line).

CONCLUSION

These results show that NeoHelix Regenerate at 1% significantly reduces the amount of damaged collagen, supporting the skin's natural collagen repair processes, and increases hyaluronic acid levels in the dermis. Compared with the placebo, the treatment appears to strengthen the dermal matrix and support the skin regeneration.

MATERIALS&METHODS

Study number
EMB230801-10172

Tested products
NeoHelix Regenerate BC10226 (Batch PA240731.01).

Study design
The clinical study was conducted in a randomized, double blind design on the forearms *vs* placebo. The efficacy of the gel cream emulsion containing 1% NeoHelix Regenerate was evaluated relative to baseline (before treatment, D0) and to the placebo gel cream emulsion (same formulation without NeoHelix Regenerate). The gel cream formulations are detailed in Annex 2. The study lasted 56 days, with evaluation points at D0 and D56.

Inclusion criteria
6 healthy female volunteers aged 60 to 70 years, phototypes I to IV, with fine skin on the inner forearms, and willing to undergo four cutaneous punch biopsies were recruited. 5 volunteers in final were included in the analysis after data quality control.

Application modality
Volunteers applied the 1% NeoHelix Regenerate gel cream emulsion or the placebo gel cream emulsion twice daily, once in the morning and once in the evening, on the left or right forearm under normal conditions of use for 56 days. The last application was performed the evening before the visit.

Evaluation method
Cutaneous biopsy protocol
Cutaneous (skin) biopsies on forearms were taken by a clinician. One 4-mm wide and 0.7-mm deep punch biopsy, unique use, was used per sampling. The size and depth of the biopsy were necessary to ensure that a sufficient amount of tissue was collected for the analysis to be conducted. A local antiseptic product was applied, then a local anesthetic by injecting lignocaine 1-2% intradermally (single use) was administered to the site prior to excision of the biopsy. Following the biopsy, a single stitch with nonabsorbable thread (size 4.0 mm) was realized and/or Steristrips were eventually applied. The site was covered with a protective dressing. Instructions on changing the dressing and spare dressings were provided. The stitch was then removed by a clinician 7-10 days later. Two cutaneous biopsies per condition (active product zone at D0, placebo product zone at D0, active product zone at D56, placebo product zone at D56) were taken. The samples were stored at -80 °C, and the biopsies were embedded in optimal cutting temperature (OCT) compound prior to staining.

Damaged collagens staining
Damaged collagens in skin biopsies were assessed using a specific fluorescent probe. Serial 10 µm tissue cryosections were fixed with acetone at -20°C for 10 minutes, then blocked with 2% serum for 2 hours at room temperature. After washing, sections were incubated with 20 µM fluorescein-conjugated collagen hybridizing peptide (F-CHP, 3Helix) for 16 hours at 4°C, prior to a 5-minute pre-heating of F-CHP solution at 80°C. The slides were then washed three times with PBS for 10 minutes each, then mounted with Vectashield Antifade Medium containing DAPI. Finally, they were analyzed using fluorescence microscopy (LEICA SP8) and image analysis.

Hyaluronic acid staining
Because of its structural simplicity and ubiquity, hyaluronic acid is non-immunogenic, making classical antibody-based detection methods ineffective. Instead, hyaluronic acid is commonly measured using a biotinylated hyaluronan-binding protein (HABP), which binds specifically to HA and enables its visualization and quantification in biological samples. Tissue cryosections (10 µm) were fixed with 4% paraformaldehyde for 20 minutes at room temperature and blocked with 2% serum for 2 hours. After washing, sections were incubated with biotinylated HABP (Amsbio) for 16 hours at 4°C, followed by Streptavidin Alexa 488. Slides were washed three times in PBS for 10 minutes each and mounted with Vectashield Antifade Medium containing DAPI for nucleus counter-staining, then analyzed using fluorescence microscopy (LEICA SP8) and image analysis.

Results and statistics
The level of damaged collagens and hyaluronic acid was quantified by image fluorescence using FIJI software (Image J2, open source).
Damaged collagens were quantified by analyzing their area within a defined region of interest, corresponding to the dermal zone extending up to 120 micrometers beneath the epidermal-dermal junction. Results are expressed as the percentage change from Day 0 in the proportion of damaged collagen area.
Hyaluronic acid was quantified by assessing its intensity within the region of interest, corresponding to the dermal zone extending up to 120 micrometers beneath the epidermal-dermal junction. Results are expressed as the percentage change from Day 0 in the intensity of Hyaluronic acid staining.
A total of 6 images were taken per donor, resulting in n= 29-30 after quality data control. The results are presented as mean value. Statistical analysis of data was performed using SigmaPlot software (V12.5) (Systat Software Inc. San Jose, California United States of America). A two-tailed test, Student t test or Mann-Whitney test, was used to compare the effect of the treatment over time and versus the placebo, with a significance threshold set at 5% ($p<0.05$).

EFFICACY

Improvement of skin aging: reducing wrinkles increasing skin tonicity and thickness

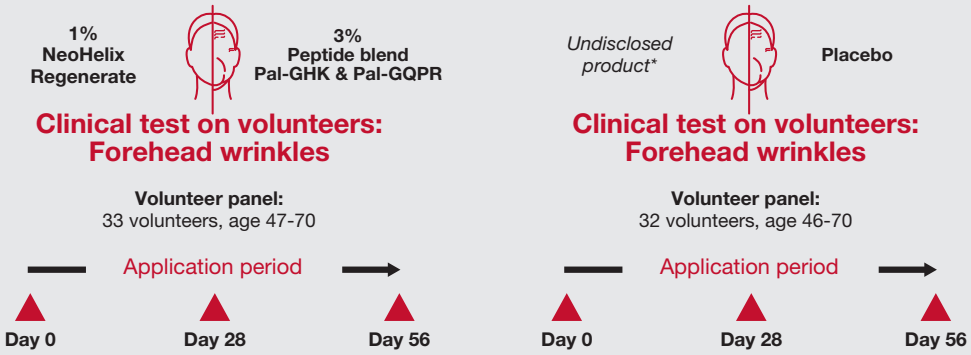
in vivo

OBJECTIVE

Among the many factors that shape overall well-being, maintaining healthy and youthful-looking skin plays a crucial role [Forte *et al*, 2015; Voegeli *et al*, 2023]. Skin aging results from a combination of intrinsic and extrinsic stressors that progressively alter dermal structure and function. Intrinsic aging mechanisms, including glycation and oxidative stress, drive progressive damage on collagen and weaken the skin's natural repair capacity. At the same time, extrinsic aggressors such as UV radiation and pollution amplify this damage, accelerating the formation of wrinkles, particularly in areas more exposed to environmental stress [Tobin, 2017; Robert *et al*, 2012].

In a randomized, split-face, double-blind clinical trial conducted on 65 volunteers aged 46-70 years old, the efficacy of NeoHelix Regenerate at 1% was evaluated in comparison with a 3% commercial peptide blend (benchmark) and a placebo formulation. The study assessed improvements in forehead wrinkle appearance together with changes in skin tonicity and thickness (Fig. 24).

The commercial peptide blend used as benchmark is a solution of ~100 ppm Palmitoyl Tripeptide-1 and ~50 ppm of Palmitoyl Tetrapeptide-7 (Pal-GHK & Pal-GQPR) (INCI: Glycerin, Water, Butylene Glycol, Carbomer, Polysorbate 20, Palmitoyl Tripeptide-1, Palmitoyl Tetrapeptide-7). It was formulated at its recommended dose of use, 3%, in the same gel cream formulation as NeoHelix Regenerate.



Randomized application, split face, twice daily for 8 weeks.
Fringe Projection (AEVA) measurement of forehead wrinkles
Cutometer measurement of skin tonicity
DUB SkinScanner measurement of skin thickness

** An undisclosed research product was tested in parallel in this clinical study; however, its results are not reported*

Figure 24 - Clinical study design.

RESULTS & DISCUSSION

Effect on wrinkles

Forehead Wrinkle Max Depth (Fig. 25):

After 28 days of application, NeoHelix Regenerate at 1% showed a significant decrease ($p<0.05$) in Forehead Wrinkle Max Depth of 13% as compared to baseline, as well as a significant decrease ($p<0.05$) of 21% as compared to the placebo. After 56 days of application, NeoHelix Regenerate at 1% showed a significant decrease ($p<0.05$) in Forehead Wrinkle Max Depth of 21% as compared to baseline, as well as a trend of decrease ($p<0.1$) of 27% as compared to placebo.

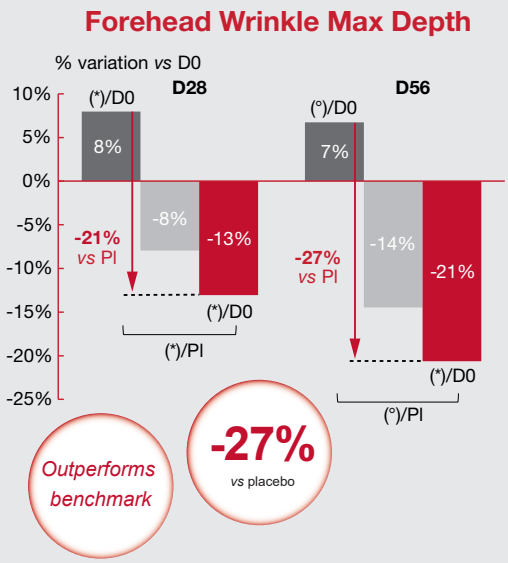
Forehead Wrinkle Mean Depth (Fig. 26):

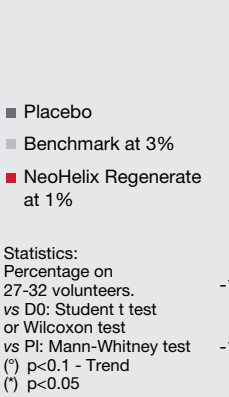
After 28 days of application, NeoHelix Regenerate at 1% also showed a trend of decrease ($p<0.1$) in Forehead Wrinkle Mean Depth of 9% as compared to baseline, as well as a significant decrease ($p<0.05$) of 13% as compared to the placebo. Additionally, after 56 days of application, NeoHelix Regenerate at 1% showed a significant decrease ($p<0.05$) in Forehead Wrinkle Mean Depth of 18% as compared to baseline, as well as a significant decrease ($p<0.05$) of 21% as compared to the placebo.

Comparison with benchmark (Fig. 25 and 26):

On both parameters, and at both time points, even if the difference did not reach the statistical threshold p value of 0.05, the results evidence that NeoHelix Regenerate at 1% outperformed the peptide blend benchmark at its recommended dose of use of 3%.

in vivo





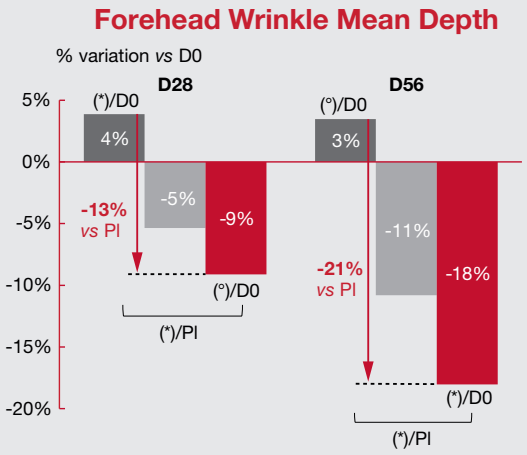


Figure 26 - Change in Forehead Wrinkle Mean Depth over time.

Figure 25 - Change in Forehead Wrinkle Max Depth over time.

These results demonstrate the ability of NeoHelix Regenerate at 1% to reduce the depth of forehead wrinkles, starting at 28 days and continuing at 56 days. This can be seen in the representative images in Fig. 27.

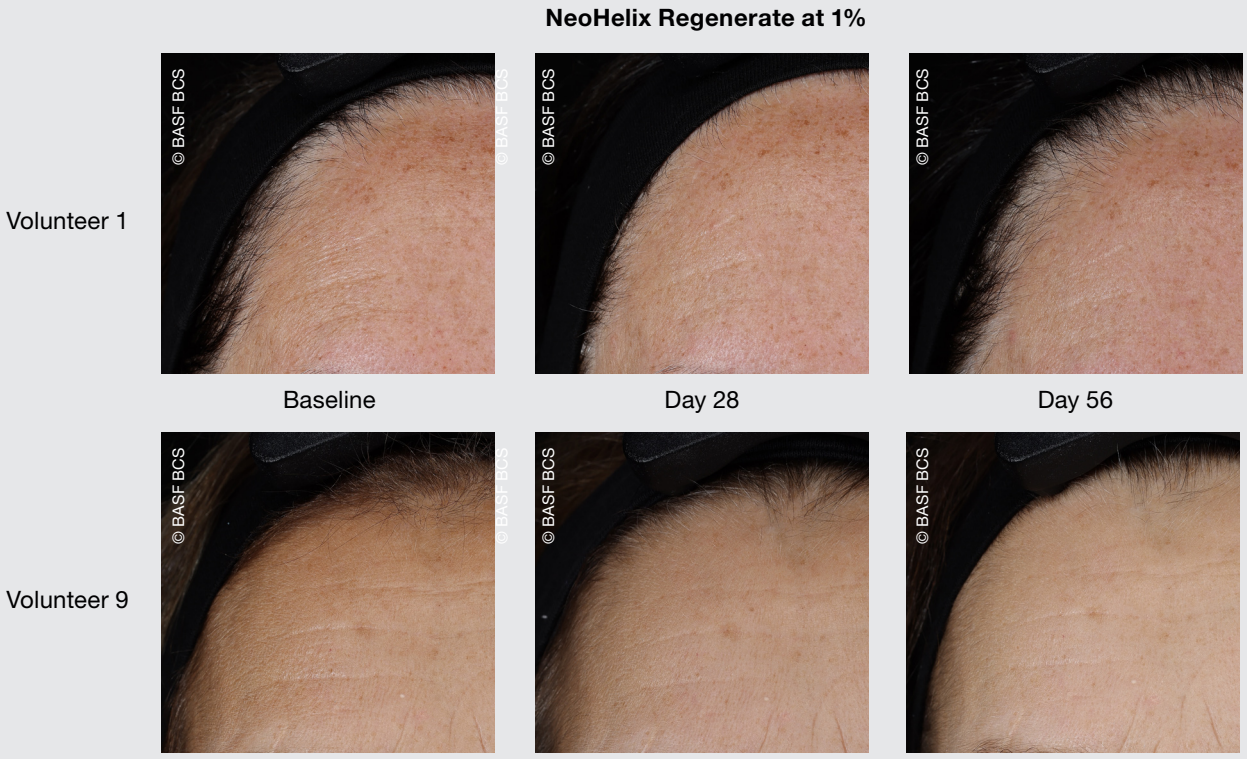


Figure 27 - Illustrative Images – Change in Forehead Wrinkles over time.

in vivo

Effect on skin tonicity (Fig. 28)

After 56 days of application, NeoHelix Regenerate at 1% showed a significant decrease ($p<0.01$) in skin viscoelasticity index (R6) of 9% as compared to baseline, as well as a significant decrease ($p<0.05$) of 10% as compared to the placebo. These results show the ability of NeoHelix Regenerate at 1% to improve skin tonicity after 56 days of application.

Compared with the peptide blend benchmark, results were similar at both time points. However, only NeoHelix Regenerate achieved statistical significance *vs* placebo ($p\leq0.05$).

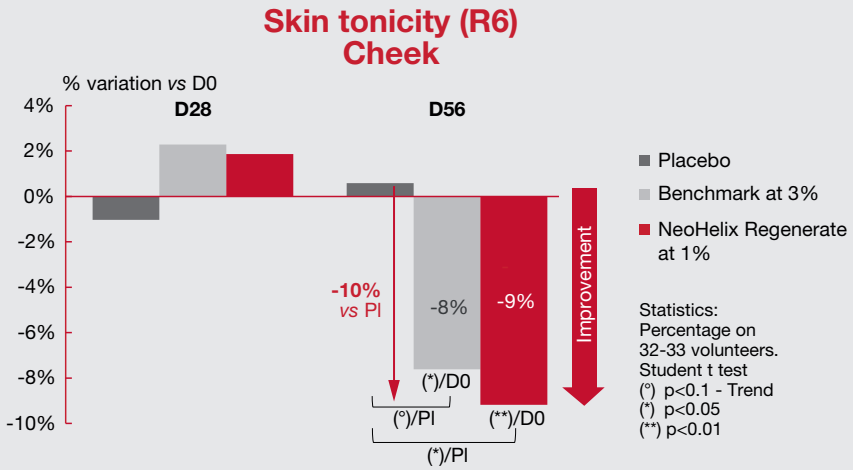


Figure 28 - Change in skin viscoelasticity (R6) over time. A decrease of R6 indicates an increase of skin tonicity.

Effect on skin thickness (Fig. 29)

After 28 days of application, NeoHelix Regenerate at 1% showed a significant increase ($p<0.05$) in skin thickness of 5% as compared to placebo (Fig. 29). After 56 days of application, NeoHelix Regenerate at 1% showed a significant increase ($p<0.05$) of 6% as compared to placebo. These results demonstrate the ability of NeoHelix Regenerate at 1% to increase skin thickness, starting at 28 days of application and continuing at 56 days of application.

Comparison with benchmark:
At D28, even if the difference did not reach the statistical threshold p value of 0.05, the result evidence that NeoHelix Regenerate at 1% outperformed the peptide blend benchmark at its recommended dose of use of 3%. After D56, the difference between the two products met the statistical significance level ($p<0.01$) with an increase of 4% over the Benchmark.

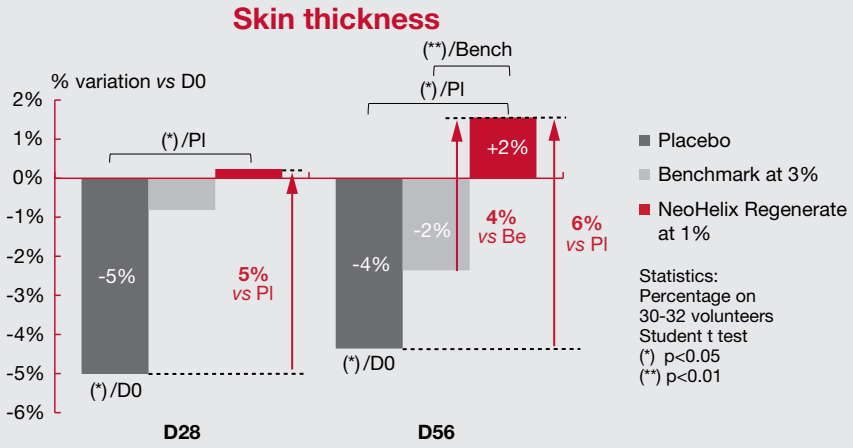


Figure 29 - Change in skin thickness over time.

CONCLUSION

NeoHelix Regenerate at 1% showed the ability to improve forehead wrinkle depth and skin thickness after only 28 days of application. This effect continues at 56 days of application, and an improvement in skin tonicity is also present after 56 days of application. Overall, NeoHelix at 1% shows anti-aging efficacy starting at 28 days of application and continuing at 56 days of application, outperforming the peptide blend benchmark (Palmitoyl Tripeptide-1 and Palmitoyl Tetrapeptide-7) at its recommended dose of use of 3%.

MATERIALS&METHODS

Study number
EMB230801-10173

Tested products
- NeoHelix Regenerate BC10226 (Batch PA240731.01) at 1%.
- Commercial peptide blend used as benchmark: solution of ~100 ppm Palmitoyl Tripeptide-1 and ~50 ppm of Palmitoyl Tetrapeptide-7 (Pal-GHK & Pal-GQPR) (INCI: Glycerin, Water, Butylene Glycol, Carbomer, Polysorbate 20, Palmitoyl Tripeptide-1, Palmitoyl Tetrapeptide-7) at 3%.

Study design
The clinical study was conducted as a double-blind, randomized, split-face trial involving two products formulated in the same gel cream formulation namely NeoHelix Regenerate at 1% and the commercial peptide blend at 3%, and a placebo gel cream, as detailed in Annex 2. Another undisclosed research product was tested in parallel in this clinical study, but its results are not disclosed in this document. The efficacy of the formulations was compared to the baseline (before treatment, D0) and between them. The study was conducted during a period of 56 days with check points at D0, D28 and D56.

Inclusion criteria
The study was done on 65 healthy female volunteers, divided in two groups 2 groups of 32-33, with Fitzpatrick skin type I-IV, aged from 46 to 70 years, having visible forehead wrinkles.

Application modality
One group evaluated the placebo gel cream formulation *vs* the undisclosed product in the same formulation (data not shown). A second group assessed the 1% NeoHelix Regenerate gel cream formulation in comparison with the peptide blend benchmark gel cream (same formulation). Volunteers applied the assigned formulations twice daily, once in the morning and once in the evening, for 56 days to one half of the face under normal conditions of use. The last application was done the evening before the visit.

Evaluation method
Measurement of forehead wrinkle depth using fringe projection (AEVA)
The AVEA is a 3D acquisition system for imaging. It has a projection unit combined with stereo imaging techniques that allow it to address wrinkle reduction. It is designed to quantify efficacy assessment for cosmetics, aesthetical and dermatology products and treatments.
Studied area: Forehead
Studied parameters: Max and mean depth of forehead wrinkles. The lower the values, the lower the forehead wrinkles appearance and the better the skin texture.

Measurement of skin tonicity by cutometer
The Cutometer dual MPA 580 is used to measure the elasticity of the upper skin layer. By using negative pressure, it mechanically deforms the skin. The measuring principal is based on the suction method. Negative pressure is created in the device, and the skin is drawn into the aperture of the probe and released again after a defined time. Inside the probe, the penetration depth is determined by a non-contact optical measuring system. This optical measuring system consists of a light source and a light receptor, as well as two prisms facing each other. The light intensity varies according to the penetration depth of the skin. The resistance of the skin to the negative pressure and its ability to return into its original position are displayed in real time during measurement as curves of penetration depth in distance (mm) *vs* time(s). This allows objective quantification of the elastic and mechanical properties of skin.
Studied area: Cheeks
Studied parameter: R6 viscoelastic parameter (Uv/Ue, Viscoelasticity index, ratio of delayed-to-immediate distension). It is increased with age, and a decrease in this parameter suggests an improvement in skin tonicity.

Measurement of skin thickness by ultrasound (DUB SkinScanner)
The DUB SkinScanner is a versatile imaging system which provides high resolution ultrasound images with excellent image contrast. In the ultrasound system, a sound wave is produced by a piezoelectric transducer, and strong, short electrical pulses from the system drive the transducer at the desired frequency. It is necessary to use ultrasound frequencies above 20 MHz to achieve very fine and high-resolution ultrasound images. High frequency ultrasound (20 to 100 MHz) uses single-element ultrasound transducers which focus the beam with physical lenses. This focusing produces an arc-shaped sound wave from the face of the transducer. The wave travels into the body and comes into focus at a desired depth. It is partially reflected from the layers between different tissues or scattered from smaller structures. The return of the sound wave vibrates the transducer, and the signal is converted to electrical pulses. The ultrasonic scanner then processes and transforms the electrical pulses into a digital image by determining the amount of time between the transmission of the sound and reception of the echo. This time difference is used to calculate the depth of the tissue interface causing the echo. The greater the difference between acoustic impedances, the larger the echo. 2 major modes of ultrasound are used in imaging: A mode (amplitude mode) and B mode (brightness mode). The X-axis of the image shows the spreading into the depth of the tissue.
Studied area: Cheeks
Studied parameter: Skin thickness. Higher values indicate a higher skin thickness.

Representative images by VISIA imaging
The VISIA has a multi-point positioning system and ghost imaging capability that captures well registered images to document progress over time.

Results and statistics
Results are expressed as the mean percentage of change compared to the baseline measurement (D0). The comparisons between products are expressed as difference between their percentage of changes over baseline.
All data have been analysed to confirm the normality of the distribution using the Shapiro-Wilk Test. The statistical analysis of the change in the studied parameters for each product over time, as well as differences in the studied parameters between the treatment groups were performed as follows:
- validation of the normality of the studied parameters: Paired t test (all values *vs* Baseline, Benchmark at 3% *vs* NeoHelix Regenerate at 1%) or t-test (Placebo *vs* Benchmark at 3%, Placebo *vs* NeoHelix Regenerate at 1%),
- invalidation of the normality of the studied parameters: Wilcoxon Signed Rank Test (all values *vs* Baseline, Benchmark at 3% *vs* NeoHelix Regenerate at 1%) or Mann-Whitney Rank Sum test (Placebo *vs* Benchmark at 3%, Placebo *vs* NeoHelix Regenerate at 1%).

All statistical tests were run at 95% confidence level (p value < 0.05).



GENERAL CONCLUSION

NeoHelix Regenerate: the smart peptide for next-level skin

Pioneering collagen-repair innovation to regenerate the skin and elevate visible anti-aging performance

- A new class of biomedical-inspired patented peptide that targets and repairs damaged collagen, restores cell-matrix communication, and reactivates fibroblasts.
- Multi-layer regenerative activity from epidermis to hypodermis.
- Clinically proven collagen repair and anti-aging performance, outperforming a leading peptide benchmark.

Among the multiple factors influencing skin quality and perceived youthfulness, the integrity of the collagen network plays a central role. As the skin is exposed to intrinsic stresses such as glycation and oxidative stress and extrinsic aggressors such as UV radiation and pollution, collagen-becomes progressively damaged. This deterioration alters the matrix quality, weakens fibroblast communication, and accelerates visible signs of aging.

Inspired by biomedical research on the Collagen Hybridizing Peptide technology, NeoHelix Regenerate introduces a new class of breakthrough peptides designed to selectively bind to damaged collagen and target skin regeneration. Its structure combines collagen-mimetic repeats with an integrin-binding motif, enabling the peptide to anchor precisely onto altered collagen fibers and restore the lost cell-matrix dialogue.

Building on this mechanism, NeoHelix Regenerate has demonstrated *in vitro* its ability to act across multiple skin layers, confirming a true multi-layer regenerative profile.

In the epidermis, it accelerates turnover while reducing inflammatory signaling by keratinocytes. In the dermis, it stimulates collagen I and III synthesis and protects the matrix from glycation, preserving structural quality. In the hypodermis, it restores adipocyte-fibroblast communication by normalizing leptin signaling from adipocytes, an essential marker of youthful deep-skin dialogue.

In vivo, NeoHelix Regenerate was tested at 1% for 56 days in two placebo-controlled clinical trials on human volunteers. It demonstrated a remarkable -41% reduction in damaged collagen in skin biopsies and a +65% increase in dermal hyaluronic acid content over baseline, statistically significant over placebo, confirming its ability to promote global skin regeneration. NeoHelix Regenerate also significantly improved multiple clinical aging signs with -27% forehead wrinkle max depth, +10% improvement in skin tonicity, and +6% increase in skin thickness over placebo, outperforming a benchmark (tri- and tetra-peptide) blend.

NeoHelix Regenerate is a new generation of smart peptide technology, combining precision-targeted collagen repair, multi-layer regeneration, and clinically proven anti-aging performance. It offers formulators a high-value, science-backed cosmetic ingredient designed to meet the growing demand for potent, targeted, and biologically intelligent skin-regeneration solutions.

ANNEXES

Annex 1 - Technical data - Available upon request

- Quality and regulatory product information
- Toxicological abstract
- Composition sheet
- Specifications
- Formulation Data Sheet
- Natural content origin (according to ISO 16128)
- Product Purity Profile

Annex 2 - Clinical test formula

Trade name	INCI name	Placebo formulation %	Peptide blend benchmark formulation %	NeoHelix Regenerate formulation %
Water	Water	qsf. 100	qsf. 100	qsf. 100
Cetiol C 5C	Coco-Caprylate/Caprate	6.50	6.50	6.50
Dimethyl Isosorbide	Dimethyl Isosorbide	2.00	2.00	2.00
Glycerine 99.5% MB	Glycerin	2.00	2.00	2.00
Dehymuls PGPH	Polyglyceryl-2 Dipolyhydroxy-stearate	2.00	2.00	2.00
Eumulgin SML 20	Polysorbate 20	2.00	2.00	2.00
Plantapon LGC Sorb	Sodium Lauryl Glucose Carboxylate, Lauryl Glucoside	1.75	1.75	1.75
Cosmedia SP	Sodium Polyacrylate	1.50	1.50	1.50
Euxyl PE 9010 (Ashland)	Phenoxyethanol, Ethylhexylglycerin	1.10	1.10	1.10
Verdessence Xanthan	Xanthan Gum	0.10	0.10	0.10
Eumulgin Prisma	Disodium Cetearyl Sulfosuccinate	0.08	0.08	0.08
Benchmark*	Glycerin, Water, Butylene Glycol, Carbomer, Polysorbate 20, Palmitoyl Tripeptide-1, Palmitoyl Tetrapeptide-7	-	3.00	-
NeoHelix Regenerate BC10226	Mannitol, Acetyl Tetra (Tripeptide-29) Dipeptide-73 Hydroxyproline Tripeptide-122 Tetra (Tripeptide-29)	-	-	1.00

* The Peptide blend formulation was only used in the second clinical test on skin aging appearance (wrinkles, skin tonicity and thickness).

BIBLIOGRAPHY

Ahmad AA, Ghim M, Kukreja G, Neishabouri A, Zhang Z, Li J, Salarian M, Toczek J, Gona K, Hedayatyanfard K, Morrison T, Zhang J, Huang YH, Liu C, Yu SM, Sadeghi MM.
Collagen Hybridizing Peptide-Based Radiotracers for Molecular Imaging of Collagen Turnover in Pulmonary Fibrosis.
J Nucl Med. 2025 Mar 3;66(3):425-433.

Bennink LL, Li Y, Kim B, Shin IJ, San BH, Zangari M, Yoon D, Yu SM.
Visualizing collagen proteolysis by peptide hybridization: From 3D cell culture to in vivo imaging.
Biomaterials. 2018 Nov; 183:67-76.

Bickel M.
The role of interleukin-8 in inflammation and mechanisms of regulation.
J of Periodontology, 1993, 64(5 Suppl), 456–460.

Chakraborty S, Banerjee S, Raina M, Haldar S.
Force-Directed "Mechanointeractome" of Talin-Integrin.
Biochemistry. 2019 Nov 26;58(47):4677-4695.

Chen C, Manso AM, Ross M.
Talin and Kindlin as Integrin-Activating Proteins: Focus on the Heart.
Pediatr Cardiol. 2019 Oct.; 40(7): 1401–1409.

Choudhary V, Choudhary M, Bollag W.B. Exploring Skin Wound Healing Models and the Impact of Natural Lipids on the Healing Process. Int. J. Mol. Sci. 2024, 25, 3790.

Converse MI, Walther RG, Ingram JT, Li Y, Yu SM, Monson KL.
Detection and characterization of molecular-level collagen damage in overstretched cerebral arteries.
Acta Biomater. 2018 Feb; 67:307-318.

Danby FW.
Nutrition and aging skin: sugar and glycation.
Clin Dermatol. 2010 Jul-Aug;28(4):409-11.

Elango J, Hou C, Bao B, Wang S, Maté Sánchez de Val JE, Wenhui W.
The Molecular Interaction of Collagen with Cell Receptors for Biological Function.
Polymers (Basel). 2022 Feb 23;14(5):876.

Emam AN.
Collagen and collagen-derived materials: synthesis, structure, classification, fundamental properties and biomedical applications.
Applied Sciences, 2025, 7:1114.

Emsley J, Knight CG, Farndale RW, Barnes MJ, Liddington RC.
Structural basis of collagen recognition by integrin $\alpha 2\beta 1$.
Cell. 2000 Mar 31;101(1):47-56.

Emsley J, Knight CG, Farndale RW, Barnes MJ.
Structure of the integrin alpha2beta1-binding collagen peptide.
J Mol Biol. 2004 Jan 23;335(4):1019-28.

Ezure T, Amano S.
Adiponectin and leptin up-regulate extracellular matrix production by dermal fibroblasts.
Biofactors. 2007;31(3-4):229-36.

Feng C, Chen X, Yin X, Jiang Y, Zhao C.
Matrix Metalloproteinases on Skin Photoaging.
J Cosmet Dermatol. 2024 Dec;23(12):3847-3862.

Fernandez-Guarino M, Naharro-Rodriguez J, Bacci S.
Aberrances of the Wound Healing Process: A Review.
Cosmetics 2024, 11, 209.

Fields GB.
Interstitial collagen catabolism.
J Biol Chem. 2013 Mar 29;288(13):8785-93.

Fisher GJ, Wang B, Cui Y, Shi M, Zhao Y, Quan T, Voorhees JJ.
Skin aging from the perspective of dermal fibroblasts: the interplay between the adaptation to the extracellular matrix microenvironment and cell autonomous processes.
J Cell Commun Signal. 2023 Sep;17(3):523-529.

Forte AJ, Andrew TW, Colasante C, Persing JA.
Perception of Age, Attractiveness, and Tiredness After Isolated and Combined Facial Subunit Aging.
Aesthetic Plast Surg. 2015 Dec;39(6):856-69.

Gillitzer R, Goebeler M.
Chemokines in cutaneous wound healing.
J Leukoc Biol. 2001 Apr;69(4):513-21.

Han AR, Nam MH, Lee KW.
Plantamajoside Inhibits UVB and Advanced Glycation End Products-Induced MMP-1 Expression by Suppressing the MAPK and NF-κB Pathways in HaCaT Cells.
Photochem Photobiol. 2016 Sep;92(5):708-19.

Huang J, Heng S, Zhang W, Liu Y, Xia T, Ji C, Zhang LJ.
Dermal extracellular matrix molecules in skin development, homeostasis, wound regeneration and diseases.
Semin Cell Dev Biol. 2022 Aug; 128:137-144.

Hwang J, Huang Y, Burwell TJ, Peterson NC, Connor J, Weiss SJ, Yu SM, Li Y.
In Situ Imaging of Tissue Remodeling with Collagen Hybridizing Peptides.
ACS Nano. 2017 Oct 24;11(10):9825-9835.

Hynes RO.
Integrins: bidirectional, allosteric signaling machines.
Cell. 2002 Sep 20;110(6):673-87.

Jadach B, Mielcarek Z, Osmałek T.
Use of Collagen in Cosmetic Products.
Curr Issues Mol Biol. 2024 Mar 4;46(3):2043-2070.

Jokinen J, White DJ, Salmela M, Huhtala M, Käpylä J, Sipilä K, Puranen JS, Nissinen L, Kankaanpää P, Marjomäki V, Hyypiä T, Johnson MS, Heino J.
Molecular mechanism of alpha2beta1 integrin interaction with human echovirus 1.
EMBO J. 2010 Jan 6;29(1):196-208.

Kanniyappan H, Chathurika Rathnayake RA, Osamor J, Islam M, Wang RR.
The role of collagen and collagen I/III ratio in pathological conditions: insights into molecular mechanisms and therapeutic approaches.
Front Bioeng Biotechnol. 2025 Oct 9; 13:1679625.

Kasper M, Funk RH.
Age-related changes in cells and tissues due to advanced glycation end products (AGEs).
Arch Gerontol Geriatr. 2001 Jun;32(3):233-43.

Khew ST, Tong YW.
Template-assembled triple-helical peptide molecules: mimicry of collagen by molecular architecture and integrin-specific cell adhesion.
Biochemistry. 2008 Jan 15;47(2):585-96.

Krishnamoorthy D, Hoy RC, Natelson DM, Torre OM, Laudier DM, Iatridis JC, Illien-Jünger S.
Dietary advanced glycation end-product consumption leads to mechanical stiffening of murine intervertebral discs.
Dis Model Mech. 2018 Dec 18;11(12): dmm036012.

Knight CG, Morton LF, Peachey AR, Tuckwell DS, Farndale RW, Barnes MJ.
The collagen-binding A-domains of integrins alpha(1)beta(1) and alpha(2)beta(1) recognize the same specific amino acid sequence, GFOGER, in native (triple-helical) collagens.
J Biol Chem. 2000 Jan 7;275(1):35-40.

Knight CG, Morton LF, Peachey AR, Tuckwell DS, Farndale RW, Barnes MJ.
The collagen-binding A-domains of integrins alpha(1)beta(1) and alpha(2)beta(1) recognize the same specific amino acid sequence, GFOGER, in native (triple-helical) collagens.
J Biol Chem. 2000 Jan 7;275(1):35-40.

Kobayashi K, Igimi H.
Glycation index of hair for non-invasive estimation of diabetic control.
Biol Pharm Bull. 1996 Apr;19(4):487-90.

Kwa KAA, van Haasterecht L, Elgersma A, Breederveld RS, Groot ML, van Zuijlen PPM, Boekema BKHL.
Effective enzymatic debridement of burn wounds depends on the denaturation status of collagen.
Wound Repair Regen. 2020 Sep;28(5):666-675.

Lai, CF, Esam G, Lin YC.
From destruction to protection: rethinking MMP-1 in skin aging and anti-aging strategies.
Arch Dermatol Res 318, 56 (2026).

Leikina E, Mertts MV, Kuznetsova N, Leikin S.
Type I collagen is thermally unstable at body temperature.
Proc Natl Acad Sci U S A. 2002 Feb 5;99(3):1314-8.

Li Y, Ho D, Meng H, Chan TR, An B, Yu H, Brodsky B, Jun AS, Michael Yu S.
Direct detection of collagenous proteins by fluorescently labeled collagen mimetic peptides.
Bioconjug Chem. 2013 Jan 16;24(1):9-16.

Li P, Jin H, Liu D, Gao L, Xu Z, Nan W, Wang Y, Gao L.
Study on the effect of leptin on fibroblast proliferation and collagen synthesis *in vitro* in rats.
Zhongguo Xiu Fu Chong Jian Wai Ke Za Zhi. 2005 Jan;19(1):20-2.

Li X, Zhang Q, Yu SM, Li Y.
The Chemistry and Biology of Collagen Hybridization.
J Am Chem Soc. 2023 May 24;145(20):10901-10916.

Linder M, Bennink L, Foxton RH, Kirkness M, Westenskow PD.
In vivo monitoring of active subretinal fibrosis in mice using collagen hybridizing peptides.
Lab Anim (NY). 2024 Aug;53(8):196-204.

Liu SQ, Tian Q, Hedrick JL, Po Hui JH, Ee PL, Yang YY.
Biomimetic hydrogels for chondrogenic differentiation of human mesenchymal stem cells to neocartilage.
Biomaterials. 2010 Oct;31(28):7298-307.

Madeira F, Madhusoodanan N, Lee J, Eusebi A, Niewielska A, Tivey ARN, Lopez R, Butcher S.
The EMBL-EBI Job Dispatcher sequence analysis tools framework in 2024.
Nucleic Acids Res. 2024 Jul 5;52(W1): W521-W525.

Mahmoud AM, Ali MM.
High Glucose and Advanced Glycation End Products Induce CD147-Mediated MMP Activity in Human Adipocytes.
Cells. 2021 Aug 16;10(8):2098.

Malcor JD, Mallein-Gerin F.
Biomaterial functionalization with triple-helical peptides for tissue engineering.
Acta Biomater. 2022 Aug; 148:1-21.

Martic I, Jansen-Dürr P, Cavinato M.
Effects of Air Pollution on Cellular Senescence and Skin Aging.
Cells. 2022 Jul 17;11(14):2220.

Pang X, He X, Qiu Z, Zhang H, Xie R, Liu Z, Gu Y, Zhao N, Xiang Q, Cui Y.
Targeting integrin pathways: mechanisms and advances in therapy.
Signal Transduct Target Ther. 2023 Jan 2;8(1):1.

Potekaev NN, Borzykh OB, Medvedev GV, Pushkin DV, Petrova MM, Petrov AV, Dmitrenko DV, Karpova EI, Demina OM, Shnayder NA.
The Role of Extracellular Matrix in Skin Wound Healing.
J Clin Med. 2021 Dec 18;10(24):5947.

Quan T, Fisher GJ.
Role of age-associated alterations of the dermal extracellular matrix microenvironment in human skin aging: a mini-review.
Gerontology. 61: 427-34 (2015)

Quan T, Little E, Quan H, Qin Z, Voorhees JJ, Fisher GJ.
Elevated matrix metalloproteinases and collagen fragmentation in photodamaged human skin: impact of altered extracellular matrix microenvironment on dermal fibroblast function.
J Invest Dermatol. 2013 May;133(5):1362-6.

Queiroz ZAJ, Velosa APP, Contini VE, Sampaio-Silva J, Catanozi S, Dos Santos Filho A, Carrasco S, Lobo TM, da Silveira LKR, Zambon Robertoni FS, Machado Baldavira C, Fernezliam SM, Santos AS, Goldenstein-Schainberg C, Sampaio-Barros PD, Levy D, Capelozzi VL, Teodoro WR.
Type V Collagen as a Critical Regulator of Fibrillar Matrix Remodeling in a Murine Model of Systemic Sclerosis.
Cells. 2025 Nov 26;14(23):1865.

Rembe JD, Garabet W, Augustin M, Dissemond J, Ibing W, Schelzig H, Stuermer EK.
Immunomarker profiling in human chronic wound swabs reveals IL-1 beta/IL-1RA and CXCL8/CXCL10 ratios as potential biomarkers for wound healing, infection status and regenerative stage.
J Transl Med. 2025 Apr 8;23(1):407.

Ridiandries A, Tan JTM, Bursill CA.
The Role of Chemokines in Wound Healing.
Int. J. Mol. Sci. 2018, 19, 3217.

Robert L, Labat-Robert J, Robert AM.
Physiology of skin aging.
Clin Plast Surg. 2012 Jan;39(1):1-8.

Salamito M, Haydont V, Pageon H, Ruggiero F, Girardeau-Hubert S.
Collagen diversity in human skin: Aging, wound healing, and disorders.
Matrix Biol. 2025.

Sander EA, El-Hattab MY, Jacobson KR, Klingelhutz AJ, Ankrum JA, Calve S.
Fibroblast-Adipocyte Lineage Cell Interactions Result in Differential Production of Extracellular Matrix Proteins.
Cell Mol Bioeng. 2024 Oct 29;17(6):597-608.

Schikowski T, Hüls A.
Air Pollution and Skin Aging.
Curr Environ Health Rep. 2020 Mar;7(1):58-64.

Schroeder AB, Karim A, Ocotl E, Dones JM, Chacko JV, Liu A, Raines RT, Gibson ALF, Eliceiri KW.
Optical imaging of collagen fiber damage to assess thermally injured human skin.
Wound Repair Regen. 2020 Nov;28(6):848-855.

Shao Y, Qin Z, Alexander Wilks J, Balimunkwe RM, Fisher GJ, Voorhees JJ, Quan T.
Physical properties of the photodamaged human skin dermis: Rougher collagen surface and stiffer/harder mechanical properties.
Exp Dermatol. 2019 Aug;28(8):914-921.

Shoulders MD, Raines RT.
Collagen structure and stability.
Annu Rev Biochem. 2009; 78:929-58.

Tobin DJ.
Introduction to skin aging.
J Tissue Viability. 2017 Feb;26(1):37-46.

Tohgasaki T, Nishizawa S, Kondo S, Ishiwatari S, Sakurai T.
Long Hanging Structure of Collagen VII Connects the Elastic Fibers and the Basement Membrane in Young Skin Tissue.
J Histochem Cytochem. 2022 Nov-Dec;70(11-12):751-757.

Tu Y, Quan T.
Oxidative Stress and Human Skin Connective Tissue Aging.
Cosmetics 2016, 3, 28.

Varani J, Dame MK, Rittie L, Fligiel SE, Kang S, Fisher GJ, Voorhees JJ.
Decreased collagen production in chronologically aged skin: roles of age-dependent alteration in fibroblast function and defective mechanical stimulation.
Am J Pathol. 2006 Jun;168(6):1861-8.

Voegeli R, Campiche R, Biassin R, Rawlings AV, Shackelford TK, Fink B.
Predictors of female age, health and attractiveness perception from skin feature analysis of digital portraits in five ethnic groups.
Int J Cosmet Sci. 2023 Oct;45(5):672-687.

Wang Y, Cai W, Kong K, Jia X, Anderson RR, Franco W
"Visualization of neo-epidermis formation and evaluation of wound closure using UV fluorescence excitation imaging at two wavelengths (Conference Presentation).
" Proc. SPIE 10851, Photonics in Dermatology and Plastic Surgery 2019, 108510H (4 March 2019).

Watt FM, Fujiwara H.
Cell-extracellular matrix interactions in normal and diseased skin.
Cold Spring Harb Perspect Biol. 2011 Apr 1;3(4): a005124.

Zhang X, Yin M, Zhang LJ. Keratin 6, 16 and 17.
Critical Barrier Alarmin Molecules in Skin Wounds and Psoriasis.
Cells. 2019 Aug 1; 8(8): 807.

Zheng W, Li H, Go Y, Chan XHF, Huang Q, Wu J.
Research Advances on the Damage Mechanism of Skin Glycation and Related Inhibitors.
Nutrients. 2022 Nov 1;14(21):4588.

Zhou CJ, Guo Y.
Mini review on collagens in normal skin and pathological scars: current understanding and future perspective.
Front Med (Lausanne). 2024 Jul 18; 11:1449597.

Europe

BASF Personal Care
and Nutrition
GmbH
CM1, Rheinpromenade 1
40789 Monheim - Germany

Phone: +49 2173 4995 0
personal-care-eu@basf.com

North America

BASF Corporation
100 Park Avenue
Florham Park
NJ 07932
USA

Phone US and Canada: +1 800 962 7831
Mexico: +1 800 723 3000 | +1 800 552 4288
personal-care-na@basf.com

South America

BASF S.A. Brazil
04538-132 Sao Paulo
Brazil

Phone: +55 11 2039 2273
personal-care-sa@basf.com

Asia Pacific

BASF East Asia Regional
Headquarters Ltd.
36/F Two TaiKoo Place, Taikoo Place
979 King's Road - Quarry Bay
Hong Kong

Phone: +852 2731 0111
personal-care-hk@basf.com

Publisher
BASF Beauty Care Solutions France SAS
3, Rue de Seichamps - 54425 Pulnoy (France)
Email: bcs-europe@basf.com
www.personalcare.basf.com

Pictures
© BASF Beauty Care Solutions France SAS. All rights reserved.
®/™ Registered trademark / trademark of BASF in several countries

Global Edition, March 2026
BR234

Although all statements and information in this publication are believed to be accurate and reliable, they are presented gratis and for guidance only, and risks and liability for results obtained by use of the products or application of the suggestions described are assumed by the user. SELLER MAKES NO WARRANTY OF ANY KIND, EITHER EXPRESS OR IMPLIED, BY FACT OR LAW, INCLUDING WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. Statements or suggestions concerning possible use of the products are made without representation or warranty that any such use is free of patent infringement and are not recommendations to infringe any patent. The user should not assume that toxicity data and safety measures are indicated or that other measures may not be required. The claims and supporting data provided in this publication have not been evaluated for compliance with any jurisdiction's regulatory requirements and the results reported may not be generally true under other conditions or in other matrices. Users must evaluate what claims and information are appropriate and comply with a jurisdiction's regulatory requirements. Recipient of this publication agrees to (1) indemnify and hold harmless each entity of the BASF organization for any and all regulatory action arising from recipient's use of any claims or information in this publication, including, but not limited to, use in advertising and finished product label claims, and (2) not present this publication as evidence of finished product claim substantiation to any regulatory authority.